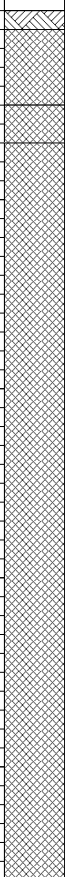
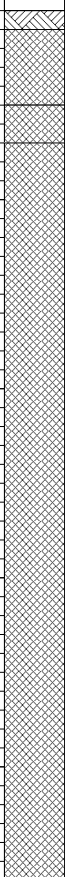
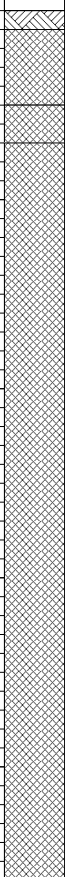
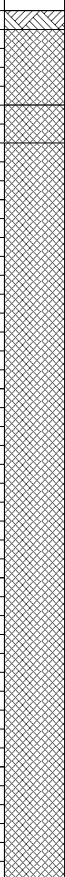
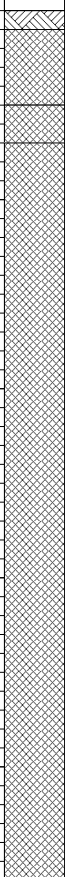
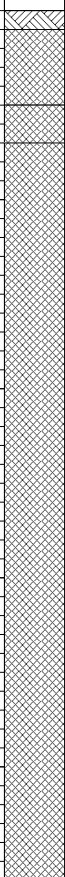


Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: V Tsambalas
Start Date: 01/12/25
Checked by: J Riley
End Date: 02/12/25

Driller: Hard Access Drilling
Drill Rig: Man Portable Drill Rig
Total Depth: 18.85 m
Hole Diameter: 100-96 mm
Inclination: 90°
Bearing: N/A
Easting: 332805.60
Northing: 6257188.00
Hor. Datum: MGA2020-56
RL: 50.96 m
Ver. Datum: AHD
Surface: Fill

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA		ES: 0.10-0.20 m PID=0.4 PPM			0.0			TOPSOIL - Silty SAND: fine to coarse grained, dark brown; trace gravel, fine to medium, sub-angular to sub-rounded ; root fibres ; topsoil.			FILL
		ES: 0.50-0.60 m PID=0.8 PPM						FILL - COBBLES: yellow-brown; sandstone.			
AD/T	HWT	ES: 1.00-1.10 m PID=0.6 PPM	Not Observed		50.0			FILL - Gravelly SAND: pale brown; Gravel, fine to coarse, sub-angular to sub-rounded sandstone ; trace roots, trace plastic fragments.		D	Potential void between large Sandstone boulders
		SPT: 1.50-1.95 m 9.19.13 N=32 D: 1.50-1.95 m ES: 1.50-1.60 m PID=1.3 PPM			49.0						
AD/T		SPT: 2.00-2.45 m 0.0.1 N=1 ES: 2.00-2.10 m PID=1.1 PPM	Not Observed								
		SPT: 3.00-3.29 m 3.20/140 mm HB N=R D: 3.00-3.29 m ES: 3.00-3.10 m PID=0.9 PPM			48.0						
AD/T		SPT: 4.50-4.55 m 10/50 mm HB N=R	Not Observed		47.0						
					46.0						
AD/T			Not Observed		45.0						
					44.0						
AD/T			Not Observed		43.0						
					42.0						

Log continued on next page.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: V Tsambalas
Start Date: 01/12/25

Checked by: J Riley
End Date: 02/12/25

Driller: Hard Access Drilling	Hole Diameter: 100-96 mm
Drill Rig: Man Portable Drill Rig	Inclination: 90°
Total Depth: 18.85 m	Bearing: N/A

Easting: 332805.60 **RL:** 50.96 m
Northing: 6257188.00 **Ver. Datum:** AHD
Hor. Datum: MGA2020-56 **Surface:** Fill

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: V Tsambalas
Start Date: 01/12/25
Checked by: J Riley
End Date: 02/12/25

Driller: Hard Access Drilling
Drill Rig: Man Portable Drill Rig
Total Depth: 18.85 m
Hole Diameter: 100-96 mm
Inclination: 90°
Bearing: N/A

Easting: 332805.60
Northing: 6257188.00
Hor. Datum: MGA2020-56
RL: 50.96 m
Ver. Datum: AHD
Surface: Fill

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □ SOL ₍₆₄₎ VL ₂ L ₄ M ₂₀ H ₆₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	3		Is(50) A=1.15 MPa Is(50) D=1 MPa				SANDSTONE: medium to coarse grained, pale grey, trace black flecks, occasional thin laminations at 0-20°.					8.00-9.01 m: HB, x 5
					42.0	9.0						
							from 9.11 to 10.63 m: becoming thickly to thinly laminated					
	9.76		C: 9.51-9.68 m Is(50) A=1.87 MPa Is(50) D=1.5 MPa		41.0	10.0			100 (98) [92]			9.51 m: HB 9.68 m: P, 20°, RF, CN 9.76 m: P, 20°, RF, CN 9.87 m: P, 10°, RF, CN 10.00 m: HB
			Is(50) A=1.51 MPa Is(50) D=1.4 MPa		40.0	11.0						10.40 m: P, 20°, RF, CN 10.49 m: P, 20°, RF, CN 10.52 m: P, 20°, RF, CT 10.73-11.57 m: DB, x 6
HQ3	4	Not Observed	Is(50) A=2.28 MPa Is(50) D=1.3 MPa		39.0	12.0	From 11.28 to 11.74 m: occasional bands of fine gravel sized quartz	FR				11.77 m: P, 5°, RF, CN 12.00 m: HB 12.25 m: P, 0°, RF, CN 12.43 m: P, 0°, RF, CN 12.50-12.55 m: SZ, 5-10°, RF, CN
			Is(50) A=0.65 MPa Is(50) D=0.42 MPa				from 12.28 m: becoming thickly to thinly laminated at 0-15° From 12.30 to 12.55 m: fine grained, grey At 12.43 m: quartz and siltstone fine gravel inclusions					12.77 m: DB 13.00 m: HB
	12.77				38.0	13.0			100 (100) [100]			
			Is(50) A=1.85 MPa Is(50) D=1.6 MPa C: 13.85-14.00 m		37.0	14.0						13.72 m: P, 5°, RF, CN 14.00 m: HB 14.21 m: P, 10°, RF, CT SC
			Is(50) A=1.43 MPa Is(50) D=1.4 MPa		36.0	15.0						15.00 m: HB
	15.86		Is(50)		35.0	16.0	SANDSTONE: medium to coarse grained, pale grey, trace black flecks, thickly to thinly laminated at 0-20°.		100			15.86 m: MB

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: V Tsambalas
Start Date: 01/12/25

Checked by: J Riley
End Date: 02/12/25

Driller: Hard Access Drilling	Hole Diameter: 100-96 mm
Drill Rig: Man Portable Drill Rig	Inclination: 90°
Total Depth: 18.85 m	Bearing: N/A

Easting: 332805.60 **RL:** 50.96 m
Northing: 6257188.00 **Ver. Datum:** AHD
Hor. Datum: MGA2020-56 **Surface:** Fill

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ ▢ ■	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	6	Not Observed	A=1.46 MPa Is(50) D=1.4 MPa Is(50) A=1.42 MPa Is(50) D=1.2 MPa Is(50) A=1.32 MPa Is(50) D=1.2 MPa Is(50) A=1.62 MPa Is(50) D=1.4 MPa	34.0 17.0 33.0 18.0 32.0 19.0 31.0 20.0 30.0 21.0 29.0 22.0 28.0 23.0 27.0			SANDSTONE: medium to coarse grained, pale grey, trace black flecks, thickly to thinly laminated at 0-20°. from 16.37 to 16.43 m: grey	FR	(100) [100]	SOIL 0-4 VL 2 L 6 M 20 H 50 VH 100 FH 200	20 60 200 600 2000	16.00 m: MB 16.38 m: P, 0° 16.57 m: P, 0° 17.00-18.60 m: MB, x 4
							Terminated at 18.85 m. Target depth.					

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: A Boot
Start Date: 27/11/25
Checked by: J Riley
End Date: 28/11/25

Driller: TerraTest
Drill Rig: Commachio Geo 305
Total Depth: 5.7 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A
Easting: 332756.10
Northing: 6257251.00
Hor. Datum: MGA2020-56
RL: 55.993 m
Ver. Datum: AHD
Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA		PID=1.3 PPM			0.0			Clayey SAND: fine to coarse grained, brown; trace roots.		TOPSOIL
		ES: 0.50-0.60 m PID=1.1 PPM						FILL - Gravelly SAND: fine to coarse grained, sub-angular to sub-rounded, grey.	M	FILL
		ES: 1.00-1.10 m PID=3.1 PPM			55.0			FILL - CLAY: orange-brown; with sand, fine.	w < PL	
		SPT: 1.50-1.95 m 4.3.4 N=7 D: 1.50-1.95 m ES: 1.50-1.60 m PID=2.3 PPM ES: 2.00-2.10 m PID=2.3 PPM			54.0			FILL - Clayey SAND: fine to coarse grained, pale grey mottled orange-brown. from 1.70 m: becoming brown-grey		
		SPT: 3.00-3.45 m 5.6.6 N=12 D: 3.00-3.45 m ES: 3.00-0.10 m PID=1.9 PPM PP=320, 370 kPa D: 3.50-4.50 m			53.0			FILL - Sandy CLAY: medium plasticity, orange-brown; Sand, fine to medium.		
		ES: 4.00-4.10 m PID=2.1 PPM			52.0			FILL - Clayey SAND: fine to medium grained, grey-brown.	M	Charcoal
		SPT: 4.50-4.95 m 4.7.6 N=13 D: 4.50-4.95 m			51.0					
		SPT: 5.44-5.44 m 5/0 mm HB N=R						CONCRETE: grey; with voids up to 5 mm; concrete (culvert).		Concrete Culvert
					50.0			Terminated at 5.70 m. Services present.		
					49.0					
					48.0					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: A Boot
Start Date: 25/11/25
Checked by: J Riley
End Date: 26/11/25

Driller: TerraTest
Drill Rig: Commachio Geo 305
Total Depth: 2.25 m
Hole Diameter: 100-99 mm
Inclination: 90°
Bearing: N/A
Easting: 332745.50
Northing: 6257255.30
Hor. Datum: MGA2020-56
RL: 55.13 m
Ver. Datum: AHD
Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA		ES: 0.10-0.20 m PID=1.2 PPM		0.0			TOPSOIL - Sandy SILT: low plasticity, brown; Sand, fine to coarse; trace clay; trace root fibres; topsoil.			TOPSOIL
	U	D: 0.40-0.50 m ES: 0.50-0.60 m PID=1.8 PPM		1.0			FILL - Clayey SAND: fine to coarse grained, orange-brown; trace gravel, fine grained, subrounded to subangular.			FILL
AD/T		SPT: 1.50-1.95 m 2.5.3 N=8 D: 1.50-1.95 m PID=1.1 PPM	Not Observed	2.0		SP	SAND: fine to coarse grained, pale grey.	M	VD	EXTREMELY WEATHERED MATERIAL
				2.25			SANDSTONE: medium to coarse grained, pale grey, moist, very low to low strength			HAWKESBURY SANDSTONE
				Terminated at 2.25 m. Target depth.						
				3.0						
				4.0						
				5.0						
				6.0						
				7.0						
				8.0						

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: A Boot
Start Date: 26/11/25
Checked by: J Riley
End Date: 28/11/25

Driller: Terratest
Drill Rig: Commachio Geo 305
Total Depth: 23.45 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A
Easting: 332738.40
Northing: 6257247.00
Hor. Datum: MGA2020-56
RL: 56.22 m
Ver. Datum: AHD
Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA		ES: 0.10-0.20 m PID=1.4 PPM		56.0	0.0			TOPSOIL - Silty SAND: fine to coarse grained, brown; trace roots; topsoil; topsoil, trace roots.	M		TOPSOIL
		ES: 0.50-0.60 m PID=0.1 PPM D: 0.60-0.70 m						FILL - Clayey SAND: fine to coarse grained, brown mottled orange-brown; trace gravel, gravel is fine, sub-angular to sub-rounded.	M		FILL
AD/T	HQ	SPT: 1.50-1.95 m 2,2,2 N=4 D: 1.50-1.95 m ES: 1.50-1.60 m PID=3.4 PPM		55.0	1.0			FILL - Silty CLAY: low to medium plasticity, dark grey, trace red-brown mottling; trace gravel, subrounded to sub angular, fine.	w > PL		
		ES: 2.50-2.60 m PID=3.1 PPM		54.0	2.0						
		SPT: 3.00-3.45 m 2,2,2 N=4 D: 3.00-3.45 m		53.0	3.0						
		ES: 3.50-3.60 m PID=2.4 PPM		52.0	4.0						
		D: 4.30-4.40 m						FILL - CLAY: medium plasticity, grey with dark grey and mottled red-brown; with sand, fine to coarse.			
		SPT: 4.50-4.64 m 12/140 mm HB N=R D: 4.50-4.55 m ES: 4.50-4.60 m PID=2.8 PPM D: 4.55-4.64 m ES: 4.70-4.80 m PID=2 PPM						CLAY: high plasticity, dark brown; trace rootlets ; organic odour.			
				51.0	5.0			Log continued on next page.			
				50.0	6.0						
				49.0	7.0						

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: A Boot
Start Date: 26/11/25

Checked by: J Riley
End Date: 28/11/25

Driller: Terratest	Hole Diameter: 96 mm
Drill Rig: Commachio Geo 305	Inclination: 90°
Total Depth: 23.45 m	Bearing: N/A

Easting: 332738.40 **RL:** 56.22 m
Northing: 6257247.00 **Ver. Datum:** AHD
Hor. Datum: MGA2020-56 **Surface:** Grass

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: A Boot
Start Date: 26/11/25
Checked by: J Riley
End Date: 28/11/25

Driller: Terratest
Drill Rig: Commachio Geo 305
Total Depth: 23.45 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A
Easting: 332738.40
Northing: 6257247.00
Hor. Datum: MGA2020-56
RL: 56.22 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)		
HQ3	4		Is(50) D=0.48 MPa Is(50) A=1.22 MPa	9.40	48.0		SANDSTONE: medium to coarse grained, pale grey, thickly to thinly laminated at 0-20°, moist, LAMINITE 60° sandstone.	FR	100 (100) [98]			8.00 m: P, SM, SN, black		
					9.0							8.98-9.00 m: EW, dark grey		
					47.0							9.40 m: MB		
					10.0							9.88 m: P, 20°, RF, SN, black 9.95 m: J, 15°, RF 10.00 m: MB		
					46.0		from 10.18 m: trace carbonaceous flecks, thickly laminated					10.20 m: P, 0°, SM		
	5		Is(50) D=0.69 MPa Is(50) A=1.23 MPa		11.0		from 10.56 m: becoming very thickly bedded					10.90 m: MB 11.00 m: MB		
					45.0									
					12.0		11.90 m: MB 12.00 m: MB							
					44.0		12.40 m: DL							
					13.0		from 13.0 to 13.20 m: very thinly bedded					13.00 m: DB		
		Is(50) D=1.11 MPa Is(50) A=1.47 MPa C: 12.00-12.40 m	12.40	43.0	100 (100) [100]				13.90 m: P, 5°, RF 14.00 m: DB					
				42.0										
				14.0										
		Is(50) D=1.41 MPa Is(50) A=1.83 MPa		41.0	100 (100) [100]				14.80 m: MB 14.90 m: P, 10°, RF 15.00 m: MB					
				41.0										
				41.0	15.40 m: DB									
		Is(50) D=1.25 MPa Is(50) A=1.02 MPa		41.0	100 (100) [100]							15.70 m: P		
				41.0										
		Is(50) D=1.31 MPa Is(50) A=1.59 MPa Is(50) D=1.2 MPa Is(50) A=1.93 MPa C: 15.40-15.67 m		41.0	100 (100) [100]									
				41.0										
		Is(50) D=1.08 MPa Is(50) A=1.47 MPa		42.0	100 (100) [100]									
				42.0										
		Is(50) D=1.47 MPa Is(50) A=1.83 MPa		43.0	100 (100) [100]									
				43.0										
		Is(50) D=1.11 MPa Is(50) A=1.47 MPa		44.0	100 (100) [100]									
				44.0										
		Is(50) D=0.69 MPa Is(50) A=1.23 MPa		46.0	100 (100) [98]									
				46.0										
		Is(50) D=0.48 MPa Is(50) A=1.22 MPa		48.0	100 (100) [98]									
				48.0										

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: A Boot
Start Date: 26/11/25

Checked by: J Riley
End Date: 28/11/25

Driller: Terratest	Hole Diameter: 96 mm
Drill Rig: Commachio Geo 305	Inclination: 90°
Total Depth: 23.45 m	Bearing: N/A

Easting: 332738.40 **RL:** 56.22 m
Northing: 6257247.00 **Ver. Datum:** AHD
Hor. Datum: MGA2020-56 **Surface:** Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ □ 200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	6		Is(50) D=1.47 MPa Is(50) A=2.03 MPa	40.0			SANDSTONE: medium to coarse grained, pale grey, very thinly bedded to thinly laminated at 0-20°.	FR	100 (100) [97]			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076
Logged by: A Boot
Start Date: 25/11/25
Checked by: P Waddell
End Date: 05/12/25

Driller: Terratest
Drill Rig: Truck Mounted Rig M-CT200
Total Depth: 9.6 m

Hole Diameter: 75 mm
Inclination: 90°
Bearing: N/A

Easting: 332747.90
Northing: 6257244.49
Hor. Datum: MGA2020-56

RL: 56.1 m
Ver. Datum: AHD
Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA		ES: 0.10-0.20 m PID=1.2 PPM		56.0	0.0			TOPSOIL - Clayey SAND: fine to coarse grained, brown; trace organics; topsoil.	M		TOPSOIL
		D: 0.40-0.60 m ES: 0.50-0.60 m PID=0.2 PPM						FILL - Silty SAND: fine to coarse grained, light brown.	M		FILL
								FILL - Silty CLAY: medium plasticity, grey mottled red; with silt; red staining.			
AD/T	U	SPT: 1.50-1.95 m 3.3.8 N=11 D: 1.50-1.95 m ES: 1.50-1.60 m PID=2.4 PPM PP=390 kPa	Not Observed	55.0	1.0						
											
											
											
DB		ES: 2.50-2.60 m PID=2.2 PPM		54.0	2.0				w > PL		
											
											
											
DB		SPT: 3.00-3.45 m 5.6.4 N=10 D: 3.00-3.45 m		53.0	3.0						
											
											
											
DB		ES: 3.50-3.60 m PID=2.9 PPM		52.0	4.0						
											
											
											
DB		SPT: 4.50-4.95 m 6.12.27 HB N=39 D: 4.50-4.95 m ES: 4.50-4.60 m PID=2.3 PPM PP=380 kPa		51.0	5.0		CI-CH	Sandy CLAY: medium to high plasticity, brown; Sand, sand is fine to coarse.	w > PL	St	RESIDUAL SOIL
								SANDSTONE: fine to coarse grained, pale grey, moist, very low strength.			EXTREMELY WEATHERED MATERIAL
								SANDSTONE: medium to coarse grained, pale grey.			HAWKESBURY SANDSTONE
											100% Water loss
DB				50.0	6.0						
											
											
											
DB				49.0	7.0						
											
											
											
DB											
											
											
											

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water

Project: Willoughby Submain Amplification Project

Location: Artarmon Reserve, Artarmon, 2064

Project No: 60764076

Logged by: A Boot

Start Date: 25/11/25

Checked by: P Waddell

End Date: 05/12/25

Driller: Terratest

Drill Rig: Truck Mounted Rig M-CT200

Total Depth: 9.6 m

Hole Diameter: 75 mm

Inclination: 90°

Bearing: N/A

Easting: 332747.90

Northing: 6257244.49

Hor. Datum: MGA2020-56

RL: 56.1 m

Ver. Datum: AHD

Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
DB	U		Not Observed	48.0				SANDSTONE: medium to coarse grained, pale grey.			
				47.0	9.0			Terminated at 9.60 m. Target depth.			
				46.0	10.0						
				45.0	11.0						
				44.0	12.0						
				43.0	13.0						
				42.0	14.0						
				41.0	15.0						
				40.0	16.0						

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Francis Street, Naremburn, 2065

Project No: 60764076
Logged by: D.Aouad
Start Date: 29/10/25

Checked by: J Riley
End Date: 05/11/25

Driller: Terratest	Hole Diameter: 96-67 mm
Drill Rig: Commachio Geo 305	Inclination: -60°
Total Depth: 54.5 m	Bearing: 175°

Easting: 332825.50
Northing: 6256744.50
Hor. Datum: MGA2020-56

RL: 83.418 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Francis Street, Naremburn, 2065

Project No: 60764076
Logged by: D.Aouad
Start Date: 29/10/25
Checked by: J Riley
End Date: 05/11/25

Driller: Terratest
Drill Rig: Commachio Geo 305
Total Depth: 54.5 m
Hole Diameter: 96-67 mm
Inclination: -60°
Bearing: 175°

Easting: 332825.50
Northing: 6256744.50
Hor. Datum: MGA2020-56
RL: 83.418 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M S H VH EH 200 600 2000	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	8.45		Is(50) A=0.91 MPa Is(50) D=0.23 MPa				LAMINITE: SHALE (50%), dark grey and grey, interlaminated with SANDSTONE (50%), fine and medium grained, pale grey; iron stained; very thinly bedded to thinly laminated at 10-30°. from 8.30 m: becoming pale grey and grey NO CORE: from 8.45 - 9.22 m.	HW - MW	58 (46) [46]			7.83-8.11 m: P, 10°, SM, PR, VN CH, spaced 20-40 mm x 4 8.19 m: CS, 10°, CT CH, 10 mm 8.29 m: P, 15°, SM, PR, VN CH 8.32 m: P, 15°, SM, PR, VN CH 8.40 m: P, 15°, SM, PR, CT CH
	6		Is(50) A=1.26 MPa Is(50) D=1.1 MPa				LAMINITE: SHALE (50%), dark grey and grey, interlaminated with SANDSTONE (50%), fine and medium grained, pale grey; iron stained; very thinly bedded to thinly laminated at 10-30°.					9.29-9.68 m: DB
	10.10		C: 10.12-10.31 m				SANDSTONE: medium grained, pale grey, occasionally interlaminated at 5-20° with SHALE, dark grey.		100 (61) [61]			HAWKESBURY SANDSTONE 10.23 m: J, 75°, PR, healed
	7		Is(50) A=1.53 MPa Is(50) D=0.39 MPa									10.81 m: P, 10°, SM, IR, CT CH 10.85 m: HB 11.00 m: MB
			Is(50) A=2.57 MPa Is(50) D=1.89 MPa									11.41-11.58 m: P, 0-5°, PR, x 3
	12.03	Not Observed	Is(50) A=1.25 MPa Is(50) D=1.45 MPa						100 (100) [100]			11.92-12.05 m: MB, x 3
	8		Is(50) A=1.82 MPa Is(50) D=1.61 MPa									
			Is(50) A=1.64 MPa Is(50) D=1.84 MPa									
							from 13.00 m-13.40 m: thinly laminated	SW - FR				12.92-16.00 m: MB, x 11
	15.35		Is(50) A=2.22 MPa Is(50) D=1.81 MPa				from 14.87 m: becoming indistinctly laminated, occasional shale inclusions		100 (100) [100]			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Francis Street, Naremburn, 2065

Project No: 60764076
Logged by: D.Aouad
Start Date: 29/10/25

Checked by: J Riley
End Date: 05/11/25

Driller: Terratest
Drill Rig: Commachio Geo 305
Total Depth: 54.5 m

Hole Diameter: 96-67 mm
Inclination: -60°
Bearing: 175°

Easting: 332825.50
Northing: 6256744.50
Hor. Datum: MGA2020-56

RL: 83.418 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • IS ₍₅₀₎ D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	9	Not Observed	Is(50) A=1.52 MPa Is(50) D=1.2 MPa	18.15	98.0	[Graphic Log]	SANDSTONE: medium grained, pale grey and grey, occasional shale inclusions.	SW - FR	100 (100) [100]	[Strength Data]	[Defect Data]	17.00 m: MB 17.14 m: P, 20°, SM, PR, CN 17.16-19.00 m: DB/ HB x 8
					17.0		16.80 m-16.87 m: shale clasts					
					18.0		LAMINITE: SHALE (50%), dark grey and grey, interlaminated with SANDSTONE (50%), fine and medium grained, pale grey; iron stained; very thinly bedded to thinly laminated at 15-30°.					
					19.0		SANDSTONE: fine to medium grained, pale grey, thinly laminated at 5-15°, occasional SHALE laminations.					
					100.0							
					19.0							
					20.0							
					20.0							
					21.0							
					21.0		LAMINITE: bedded at 30°, SHALE (50%), dark grey and grey, interlaminated with SANDSTONE (50%), medium grained, pale grey; iron stained; bedded at 30°.					
HQ3	10	Not Observed	Is(50) A=1.47 MPa Is(50) D=1.18 MPa	21.30	101.0	[Graphic Log]		FR	100 (100) [100]	[Strength Data]	[Defect Data]	20.07-20.09 m: P, 15-20°, SM, PR, VN CH, x 2 20.16-21.30 m: MB, x 12
					102.0							
					22.0							
					23.0							
					24.0							
					21.0							
					22.0							
					23.0							
					24.0							
					21.0							
HQ3	11	Not Observed	Is(50) A=1.38 MPa Is(50) D=1.39 MPa	21.30	103.0	[Graphic Log]		FR	100 (100) [100]	[Strength Data]	[Defect Data]	21.37 m: P, 15°, SM, PR, VN CH 21.72 m: MB 21.88 m: P, 15°, SM, PR, CN 21.94-24.00 m: MB, x 5
					104.0							
					22.0							
					23.0							
					24.0							
					21.0							
					22.0							
					23.0							
					24.0							
					21.0							
HQ3	11	Not Observed	Is(50) A=1.59 MPa Is(50) D=1.03 MPa	21.30	104.0	[Graphic Log]		FR	100 (100) [100]	[Strength Data]	[Defect Data]	
					105.0							
					22.0							
					23.0							
					24.0							
					21.0							
					22.0							
					23.0							
					24.0							
					21.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Francis Street, Naremburn, 2065

Project No: 60764076
Logged by: D.Aouad
Start Date: 29/10/25
Checked by: J Riley
End Date: 05/11/25

Driller: Terratest
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Total Depth: 54.5 m

Hole Diameter: 96-67 mm
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Easting: 332825.50
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Hor. Datum: MGA2020-56

RL: 83.418 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	SOIL ¹⁻⁴ VL ² L ⁶ M ²⁰ H ³⁰ VH ⁶⁰ EH ³⁰⁰	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)																			
HQ3	24.33	Not Observed	Is(50) A=1.91 MPa Is(50) D=1.25 MPa	105.0	25.0		SANDSTONE: fine to medium grained, speckled grey and dark grey, pale grey, thinly laminated at 5-15°.	FR	100 (100) [100]			24.04-27.52 m: MB, x 10																				
													Is(50) A=2.04 MPa Is(50) D=1.56 MPa	106.0	26.0	100 (100) [100]	27.78 m: P, 10°, SM, IR, CT CH 28.00-29.56 m: MB, x 6															
																		Is(50) A=1.64 MPa Is(50) D=1.81 MPa	107.0	27.0	100 (100) [100]	29.84 m: P, 15°, SM, IR, CN 30.00 m: MB										
																							Is(50) A=2.08 MPa Is(50) D=1.48 MPa	108.0	28.0	100 (100) [100]	30.34 m: P, 10°, SM, PR, VN CH 30.38-31.72 m: HB, x 7					
																												Is(50) A=1.96 MPa Is(50) D=2.23 MPa	109.0	29.0	100 (100) [100]	31.73 m: P, 30°, PR, healed 31.90-32.00 m: MB, x 3
	Is(50) A=1.61 MPa Is(50) D=0.82 MPa		111.0	31.0	100 (100) [100]				31.90-32.00 m: MB, x 3																							
										Is(50) A=2.1 MPa Is(50) D=1.88 MPa	112.0	32.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=2.04 MPa Is(50) D=1.56 MPa	113.0	33.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=1.91 MPa Is(50) D=1.25 MPa	114.0	34.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=1.78 MPa Is(50) D=1.12 MPa	115.0	35.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
																														Is(50) A=1.65 MPa Is(50) D=0.99 MPa	116.0	36.0
	Is(50) A=1.52 MPa Is(50) D=0.86 MPa		117.0	37.0	100 (100) [100]				31.90-32.00 m: MB, x 3																							
										Is(50) A=1.39 MPa Is(50) D=0.73 MPa	118.0	38.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=1.26 MPa Is(50) D=0.60 MPa	119.0	39.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=1.13 MPa Is(50) D=0.47 MPa	120.0	40.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=1.00 MPa Is(50) D=0.34 MPa	121.0	41.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
																														Is(50) A=0.87 MPa Is(50) D=0.21 MPa	122.0	42.0
	Is(50) A=0.74 MPa Is(50) D=0.08 MPa		123.0	43.0	100 (100) [100]				31.90-32.00 m: MB, x 3																							
										Is(50) A=0.61 MPa Is(50) D=0.05 MPa	124.0	44.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.48 MPa Is(50) D=0.02 MPa	125.0	45.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.35 MPa Is(50) D=0.00 MPa	126.0	46.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.22 MPa Is(50) D=0.00 MPa	127.0	47.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
																														Is(50) A=0.09 MPa Is(50) D=0.00 MPa	128.0	48.0
	Is(50) A=0.00 MPa Is(50) D=0.00 MPa		129.0	49.0	100 (100) [100]				31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	130.0	50.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	131.0	51.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	132.0	52.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	133.0	53.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
																														Is(50) A=0.00 MPa Is(50) D=0.00 MPa	134.0	54.0
	Is(50) A=0.00 MPa Is(50) D=0.00 MPa		135.0	55.0	100 (100) [100]				31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	136.0	56.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	137.0	57.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	138.0	58.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	139.0	59.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
																														Is(50) A=0.00 MPa Is(50) D=0.00 MPa	140.0	60.0
	Is(50) A=0.00 MPa Is(50) D=0.00 MPa		141.0	61.0	100 (100) [100]				31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	142.0	62.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	143.0	63.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	144.0	64.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	145.0	65.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
																														Is(50) A=0.00 MPa Is(50) D=0.00 MPa	146.0	66.0
	Is(50) A=0.00 MPa Is(50) D=0.00 MPa		147.0	67.0	100 (100) [100]				31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	148.0	68.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	149.0	69.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	150.0	70.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	151.0	71.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
																														Is(50) A=0.00 MPa Is(50) D=0.00 MPa	152.0	72.0
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	153.0	73.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	154.0	74.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	155.0	75.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	156.0	76.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	157.0	77.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	158.0	78.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	159.0	79.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	160.0	80.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	161.0	81.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	162.0	82.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	163.0	83.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	164.0	84.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	165.0	85.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	166.0	86.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	167.0	87.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	168.0	88.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	169.0	89.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	170.0	90.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	171.0	91.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	172.0	92.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	173.0	93.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	174.0	94.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	175.0	95.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	176.0	96.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	177.0	97.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	178.0	98.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	179.0	99.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	180.0	100.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	181.0	101.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	182.0	102.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	183.0	103.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	184.0	104.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	185.0	105.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	186.0	106.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	187.0	107.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	188.0	108.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	189.0	109.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	190.0	110.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	191.0	111.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	192.0	112.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	193.0	113.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	194.0	114.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	195.0	115.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	196.0	116.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	197.0	117.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	198.0	118.0	100 (100) [100]	31.90-32.00 m: MB, x 3													
																				Is(50) A=0.00 MPa Is(50) D=0.00 MPa	199.0	119.0	100 (100) [100]	31.90-32.00 m: MB, x 3								
																									Is(50) A=0.00 MPa Is(50) D=0.00 MPa	200.0	120.0	100 (100) [100]	31.90-32.00 m: MB, x 3			
Is(50) A=0.00 MPa Is(50) D=0.00 MPa	201.0	121.0	100 (100) [100]	31.90-32.00 m: MB, x 3																												
					Is(50) A=0.00 MPa Is(50) D=0.00 MPa	202.0	122.0	100 (100) [100]	31.90-32.00 m: MB, x 3																							
										Is(50) A=0.00 MPa Is(50) D=0.00 MPa	203.0	123.0	100 (100) [100]	31.90-32.00 m: MB, x 3																		
															Is(50) A=0.00 MPa Is(50) D=0.00 MPa	204.0	124.0	100 (100) [100]														

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Francis Street, Naremburn, 2065

Project No: 60764076
Logged by: D.Aouad
Start Date: 29/10/25
Checked by: J Riley
End Date: 05/11/25

Driller: Terratest
Drill Rig: Commachio Geo 305
Total Depth: 54.5 m
Hole Diameter: 96-67 mm
Inclination: -60°
Bearing: 175°

Easting: 332825.50
Northing: 6256744.50
Hor. Datum: MGA2020-56
RL: 83.418 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • IS ₍₅₀₎ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	33.40	Not Observed	Is(50) A=1.72 MPa Is(50) D=1.44 MPa	112.0	33.0		SANDSTONE: fine to coarse grained, pale grey, indistinctly laminated at 5-30°, occasional SHALE inclusions.					32.10 m: HB 32.17 m: MB
			Is(50) A=1.15 MPa Is(50) D=1.07 MPa	113.0	34.0		from 33.00 m - 33.40m: shale clasts		100 (100) [100]			32.93 m: MB 32.98 m: P, 25°, SM, PR, CT CH 33.00 m: DB
			Is(50) A=0.88 MPa Is(50) D=0.95 MPa	114.0	35.0		at 33.40 m: shale clast 60 mm thick					33.40-35.00 m: DB, x 5
			Is(50) A=1.71 MPa Is(50) D=0.58 MPa	115.0	36.0		from 33.40 m - 34.00 m: cross-bedding					
			Is(50) A=1.66 MPa Is(50) D=1.39 MPa	116.0	37.0							
			Is(50) A=1.34 MPa Is(50) D=1.29 MPa	117.0	38.0		at 35.65: shale clast 40 mm thick					35.67 m: P, 10°, SM, PR, CT CH
			Is(50) A=1.61 MPa Is(50) D=1.54 MPa	118.0	39.0				100 (100) [100]			35.86 m: HB 35.86-38.00 m: MB, x 10
			Is(50) A=1.99 MPa Is(50) D=1.51 MPa	119.0	40.0							38.11 m: P, 10°, SM, UN, CT CH 38.26-40.00 m: MB, x 4

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Francis Street, Naremburn, 2065

Project No: 60764076
Logged by: D.Aouad
Start Date: 29/10/25
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Driller: Terratest
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Total Depth: 54.5 m
Hole Diameter: 96-67 mm
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Bearing: 175°
Easting: 332825.50
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Hor. Datum: MGA2020-56
RL: 83.418 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)	
HQ3	17	Not Observed	Is(50) A=2.41 MPa Is(50) D=2.01 MPa	41.0		SANDSTONE: fine to coarse grained, pale grey, indistinctly laminated at 5-30°, occasional SHALE inclusions.	FR	100 (100) [100]			40.00-42.00 m: MB, x 3		
			C: 41.17-41.46 m										
			Is(50) A=2.17 MPa Is(50) D=1.98 MPa	42.0									
			Is(50) A=2.69 MPa Is(50) D=2.48 MPa	43.0								42.44 m: J, 65°, RF, PR, CN, 0.1 m persistence 42.50-42.56 m: MB	
			Is(50) A=1.93 MPa Is(50) D=1.9 MPa	44.0								43.00-46.38 m: MB, x 11	
	19		Is(50) A=1.99 MPa Is(50) D=1.98 MPa	45.0					100 (100) [100]				
			Is(50) A=2.32 MPa Is(50) D=1.88 MPa	46.0									
			Is(50) A=2.35 MPa Is(50) D=1.78 MPa	47.0								46.73 m: P, 20°, SM, PR, VN CH	
												47.00-48.00 m: MB, x 5	
			Is(50) A=1.82 MPa Is(50) D=1.75 MPa										
20													

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Francis Street, Naremburn, 2065

Project No: 60764076
Logged by: D.Aouad
Start Date: 29/10/25
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Driller: Terratest
Drill Rig: Commachio Geo 305
Total Depth: 54.5 m
Hole Diameter: 96-67 mm
Inclination: -60°
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Easting: 332825.50
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Hor. Datum: MGA2020-56
RL: 83.418 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

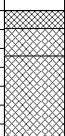
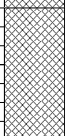
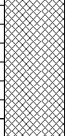
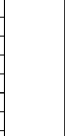
METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □ Soil: VL L M H VH EH 2 4 6 8 10 200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	48.64	Not Observed	Is(50) A=1.3 MPa Is(50) D=1.14 MPa C: 49.00-49.45 m	125.0			SANDSTONE: fine to coarse grained, pale grey, indistinctly laminated at 5-30°, occasional SHALE inclusions.		100 (100) [100]			48.48-48.49 m: P, 30°, SM, PR, CT CH, x 2 48.64-50.73 m: MB, x 4
	21		Is(50) A=1.85 MPa Is(50) D=1.77 MPa C: 49.00-49.45 m	49.0								
			Is(50) A=1.15 MPa Is(50) D=0.89 MPa	126.0								
			Is(50) A=0.71 MPa Is(50) D=0.67 MPa	50.0								
			Is(50) A=1.31 MPa Is(50) D=0.59 MPa C: 52.75-53.00 m	127.0								
	51.54		Is(50) A=0.71 MPa Is(50) D=0.67 MPa	51.0			CONGLOMERATE: fine to coarse grained, pale grey speckled dark grey, grey and grey-white, indistinct bedding, sandstone matrix.	FR	100 (100) [100]			50.89 m: P, 30°, SM, PR, VN Qz 50.95-51.18 m: MB, x 3
			Is(50) A=1.5 MPa Is(50) D=0.81 MPa	128.0			SHALE: dark grey, very thinly bedded to thickly laminated at 0-25°, with SANDSTONE laminations up to 50 mm thick.					51.23-51.25 m: P, 30°, SM, PR, CT CH, x 2 51.23-53.75 m: MB, x 10
			Is(50) A=1.26 MPa Is(50) D=0.48 MPa	52.0								
	22		Is(50) A=1.5 MPa Is(50) D=0.81 MPa	129.0			LAMINITE: INTERBEDDED SHALE (50%) & SANDSTONE (50%): SHALE is dark grey, SANDSTONE is medium grained, pale grey and grey; very thickly bedded to thinly laminated at 5-30°.					
			Is(50) A=1.26 MPa Is(50) D=0.48 MPa	53.0								
			Is(50) A=1.26 MPa Is(50) D=0.48 MPa	130.0								53.82 m: P, 15°, SM, PR, CT
			Is(50) A=1.26 MPa Is(50) D=0.48 MPa	54.0			SANDSTONE: medium grained, pale grey and grey, bedded, occasional SHALE laminations.					54.00 m: MB 54.11 m: P, 15°, SM, PR, CT 54.18-54.50 m: MB, x 3
			Is(50) A=1.26 MPa Is(50) D=0.48 MPa	55.0			Terminated at 54.50 m. Target depth.					54.40 m: MB
			Is(50) A=1.26 MPa Is(50) D=0.48 MPa	56.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Evans Lane, St. Leonards, 2065

Project No: 60764076
Logged by: V.Tsambalas
Start Date: 07/11/25
Checked by: J.Riley
End Date: 11/11/25

Driller: Terratest
Drill Rig: Comaccio Geo 205
Total Depth: 40.12 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 332911.30
Northing: 6256230.30
Hor. Datum: MGA2020-56
RL: 67.824 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
VE	HWT	ES: 0.20-0.30 m PID=4.8 PPM	Not Observed	0.0			ASPHALTIC CONCRETE: dark grey, aggregates up to 10 mm, 0.095m thick; road surface.	M		FILL Wooden sleeper visible in hole wall
		ES: 1.00-1.10 m PID=0.5 PPM		0.7			FILL - Gravelly SAND: medium to coarse grained, dark grey; Gravel, fine to coarse, angular to sub-angular, igneous. FILL - Sandy Cobble/Bouldery GRAVEL: fine to coarse grained, angular, dark brown and dark grey, igneous; Sand, fine to coarse grained; Cobbles/Boulders, up to 50 mm, angular, igneous.			
AD/T	HWT	SPT: 1.50-1.95 m 9.7.2 N=9 D: 1.50-1.95 m PID=0.3 PPM	Not Observed	1.0			FILL - Clayey SAND: medium to coarse grained, brown mottled dark grey; with gravel, fine to medium, angular to sub-angular, igneous ; with cobbles/boulders, igneous, sand, concrete.	M		
				2.0			at 1.50 m: concrete cobbles, up to 100 mm			
			Not Observed	2.5		CL	Sandy CLAY: low plasticity, pale pink-grey and yellow-brown.	w < PL	St	RESIDUAL SOIL At 2.5m: Driller notes increase in drilling resistance
				3.0						
		SPT: 3.00-3.06 m 30/60 mm HB N=R D: 3.00-3.06 m PID=0.1 PPM	Not Observed	3.0		SP	SAND: medium to coarse grained, pale pink-grey. <i>Log continued on next page.</i>	D	VD	EXTREMELY WEATHERED MATERIAL
				4.0						
			Not Observed	5.0						
				6.0						
			Not Observed	7.0						
				8.0						

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Project No: 60764076
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Start Date: 07/11/25
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Driller: Terratest
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Total Depth: 40.12 m
Hole Diameter: 115-96 mm
Inclination: 90°
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Easting: 332911.30
Northing: 6256230.30
Hor. Datum: MGA2020-56
RL: 67.824 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ L ₂ L ₄ M ₂₀ H ₆₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					67.0							
					1.0							
					66.0							
					2.0							
					65.0							
					3.0							
							<i>Log continued from previous page.</i>					
	1						NO CORE: 3.06 - 3.07 m.	MW	97 (94)			HAWKESBURY SANDSTONE
	3.40						SANDSTONE: medium to coarse grained, pale grey, trace black flecks, rare thin beds at 10-20°, faint, at 3.07 m: becoming brown-grey and pinkish-red, laminated at 10°, iron stained		[71]			3.15-3.16 m: EW, 5°, RF, PR, SP, 10 mm aperture
							at 3.11 m: becoming pale grey, trace black flecks, occasional thin laminations at 10-20°	SW	99 (95)			3.18 m: DB
							at 3.45 m: silty CLAY inclusion, low plasticity, pale grey		[94]			3.40 m: DB
					64.0							3.46-3.49 m: EW, 25°, SM, PR, CL, 30 mm aperture
					4.0							3.90 m: J, 30°, RF, PR, CN
	2											4.00 m: HB
							NO CORE: 4.20 - 4.22 m.					
							SANDSTONE: medium to coarse grained, pale grey, trace black flecks, occasional thin laminations at 10-20°.					4.48-5.08 m: DB, x 5
					63.0				89 (74)			
	4.90							FR	[73]			
					5.0							
							from 5.62 m: becoming coarse grained, occasional carbonaceous laminations at 15-20°, variable spacing.					
							at 5.79 m: SILTSTONE inclusion, dark grey, angular, coarse gravel sized	HW				5.76 m: J, 10°, RF, PR, CN
					62.0		from 5.83 m: becoming fine and medium grained, pale grey to grey, medium bedded at 5-15° with shale laminations	SW				5.78 m: J, 10°, RF, PR, CN
					6.0							5.80-5.84 m: CS, RF, CL - GC
							NO CORE: 6.24 - 6.40 m.		86 (71)			5.88 m: P, 15°, RF, PR, CT CL
							SANDSTONE: fine to medium grained, pale grey to grey, medium bedded at 5-15° with black and grey laminations, cross-bedding visible.	FR	[70]			6.00-6.03 m: SZ, 10°, RF, PR, CN
					61.0							6.16-6.69 m: P, 0-5°, RF, PR, CN, x 6
					7.0							6.76-6.78 m: CS, RF, GC
							NO CORE: 7.23 - 7.44 m.					6.79 m: DB
							SANDSTONE: medium grained, pale grey, very thinly cross-bedded at 15-30°, occasional carbonaceous laminations.	FR				6.97 m: HB
					60.0							7.00 m: HB
												7.11 m: J, 15°, RF, PR, CN
												7.23 m: DB
												7.44 m: DB
												7.68 m: P, 10°, RF, PR, CN
												7.71 m: J, 15°, RF, PR, CT CL
												7.75 m: J, 5°, RF, ST, CN
	7.90				8.0				98			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Evans Lane, St. Leonards, 2065

Project No: 60764076
Logged by: V.Tsambalas
Start Date: 07/11/25
Checked by: J.Riley
End Date: 11/11/25

Driller: Terratest
Drill Rig: Comaccio Geo 205
Total Depth: 40.12 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 332911.30
Northing: 6256230.30
Hor. Datum: MGA2020-56
RL: 67.824 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ SOLIDITY: L, M, H, V, EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)		
HQ3	5	Not Observed	Is(50) A=1.16 MPa Is(50) D=0.96 MPa				SANDSTONE: medium grained, pale grey and grey, very thinly cross-bedded at 15-30°, occasional carbonaceous laminations.	FR	(87) [63]			Driller notes 10% water loss (~50L) at 8.6 m 7.90 m: J, 20°, RF, PR, CN 8.00 m: HB 8.05-8.06 m: P, 15°, RF, PR, CN, x 2 8.12-8.17 m: CS, RF, CN 8.36-8.37 m: J, 0°, RF, PR, VN, x 2 from 8.4 m all drilling runs lose approximately ~70L 8.73-8.81 m: EW, SP - GP 8.94 m: P, 5°, RF, PR, CN 9.00 m: HB 9.03 m: HB 9.35 m: P, 15°, RF, PR, CN 9.40-12.00 m: DB, x 16 12.04 m: P, 10°, RF, PR, CN 12.40 m: P, 10°, SM, PR, CN 12.47 m: DB 12.96-13.26 m: HB, x 4 13.88-13.90 m: DB 14.00-15.00 m: HB, x 4 15.35-15.39 m: DB 15.94 m: P, 7°, SM, PR, CT CL		
				59.0	9.0		NO CORE: 8.70 - 8.73 m. SANDSTONE: medium grained, pale grey and grey, very thinly cross-bedded at 15-30°, occasional carbonaceous laminaions.	SW						
	6		Is(50) A=1.3 MPa Is(50) D=1.12 MPa						100 (100) [95]					
				58.0	10.0		from 10.00 m: becoming medium grained, very thinly bedded at 0-10°							
	7		Is(50) A=1.17 MPa Is(50) D=0.99 MPa						100 (100) [100]					
				57.0	11.0		from 11.00 to 12.88 m: cross-bedding visible							
	8		Is(50) A=1.32 MPa Is(50) D=1.1 MPa						100 (100) [100]					
				56.0	12.0									
	9		C: 12.03-12.28 m Is(50) A=1.06 MPa Is(50) D=0.28 MPa						100 (100) [100]					
				55.0	13.0		from 12.88 m: no carbonaceous laminations							
		Is(50) A=1.31 MPa Is(50) D=1.1 MPa						100 (100) [100]						
			54.0	14.0										
		Is(50) A=1.44 MPa Is(50) D=1.25 MPa						100 (100) [100]						
			53.0	15.0										
		Is(50) A=1.49 MPa Is(50) D=1.27 MPa						100 (100) [100]						
			52.0	16.0		from 15.83 m: occasional carbonaceous laminations at 0-10°, cross bedding visible								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

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Project: Willoughby Submain Amplification Project
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Project No: 60764076
Logged by: V.Tsambalas
Start Date: 07/11/25

Checked by: J.Riley
End Date: 11/11/25

Driller: Terratest	Hole Diameter: 115-96 mm
Drill Rig: Comaccio Geo 205	Inclination: 90°
Total Depth: 40.12 m	Bearing: N/A

Easting: 332911.30 **RL:** 67.824 m
Northing: 6256230.30 **Ver. Datum:** AHD
Hor. Datum: MGA2020-56 **Surface:** Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) RQD (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ Soil 0-4 VL 2 L 6 M 8 R 10 S 12 T 14 U 16 V 18 W 20 L H VH EH	DEFECT SPACING (mm) 20 60 200 600 2000	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	10		Is(50) A=1.27 MPa Is(50) D=1.26 MPa	16.90	51.0		SANDSTONE: medium grained, pale grey and grey, very thinly cross-bedded at 15-30°, occasional carbonaceous laminaions. from 16.4 m: no carbonaceous laminations					16.00 m: HB 16.08 m: P, 15°, SM, PR, CT CL 16.41 m: P, 5°, RF, PR, CN
	11		Is(50) A=2.25 MPa Is(50) D=1.42 MPa	18.40	17.0		from 17.11 to 17.12 m: SILTSTONE inclusion, dark grey, angular at 17.25 m: becoming medium grained, pale grey, cross-bedded at 5-30°, occasional carbonaceous laminations, variably spaced	FR	100 (100) [100]			16.90 m: HB 17.00 m: HB 17.12 m: J, 10°, RF, IR, CN 17.26 m: P, 5°, RF, PR, CN 17.53 m: P, 5°, SM, PR, VN
	12		Is(50) A=2.73 MPa Is(50) D=1.97 MPa		50.0							17.83 m: P, 15°, RF, IR, CN 17.97 m: HB 18.00 m: HB 18.21 m: P, 10°, RF, PR, CN
	13	Not Observed	Is(50) A=2.53 MPa Is(50) D=1.4 MPa	19.90	18.0				100 (92) [51]			18.40 m: J, 10°, RF, PR, CT CL 18.48 m: J, 5°, RF, ST 18.60 m: P, 13°, RF, PR, CN 18.63 m: P, 10°, SM, PR, CT CL 18.68-18.71 m: EW, GP Quartz 18.77 m: J, 30°, RF, PR, CN 18.80 m: DB 18.83 m: DB 19.00 m: HB 19.04 m: J, 15°, RF, PR, CN 19.23 m: J, 25°, RF, PR, CN 19.40 19.45-19.48 m: CS, 5-15°, RF, PR, CN 19.65 m: P, 0°, RF, IR, CN 19.67 m: J, 15°, SM, PR, CN 19.76-19.82 m: EW, GP 19.90 m: P, 0°, RF, PR, CN 20.00 m: HB
	14		Is(50) A=1.66 MPa Is(50) D=1.13 MPa		49.0		CONGLOMERATE: fine - coarse grained, pale grey speckled dark grey, indistinct bedding, sandstone matrix. SANDSTONE: medium grained, pale grey, indistinct bedding, occasional carbonaceous flecks.					20.24 m: DB
	15		Is(50) A=1.66 MPa Is(50) D=1.13 MPa	21.40	48.0		SHALE: dark grey, laminated at 0-5°. from 19.76 to 19.79 m: SANDSTONE inclusion, medium to coarse grained, pale grey, sub-rounded SANDSTONE: medium grained, pale grey, very thinly cross-bedded at 5-20°, occasional carbonaceous flecks. from 20.24 to 21.82 m: indistinct bedding	MW	100 (100) [100]			21.00 m: HB 21.38-21.40 m: DB 21.72-23.15 m: DB, x 8
	16		Is(50) A=1.4 MPa Is(50) D=1.21 MPa	22.90	47.0			FR	100 (100) [100]			23.52-23.54 m: EW, 0°, RF, PR, SP 23.78 m: P, 5°, RF, CU, CN 23.82 m: HB
	17		Is(50) A=1.38 MPa Is(50) D=1.15 MPa		21.0							
	18		Is(50) A=0.68 MPa Is(50) D=0.49 MPa		23.0				100 (99) [99]			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Evans Lane, St. Leonards, 2065

Project No: 60764076
Logged by: V.Tsambalas
Start Date: 07/11/25
Checked by: J.Riley
End Date: 11/11/25

Driller: Terratest
Drill Rig: Comaccio Geo 205
Total Depth: 40.12 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A

Easting: 332911.30
Northing: 6256230.30
Hor. Datum: MGA2020-56
RL: 67.824 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: □ UCS: □ L 5 M 10 H 20 V 40 EH 200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	16	Not Observed	Is(50) A=1.21 MPa Is(50) D=1.01 MPa	24.40			SANDSTONE: medium grained, pale grey, very thinly cross-bedded at 5-20°, occasional carbonaceous flecks.	FR	100 (100) [92]			24.00 m: MB 24.09 m: P, 7°, RF, PR, CN
				43.0							24.43 m: J, 22°, RF, PR, CN 24.48	
	17		Is(50) A=1.09 MPa Is(50) D=0.87 MPa	25.90			SHALE: dark grey and grey, very thinly laminated at 0-5°.		100 (100) [100]			24.69 m: P, 0°, RF, PR, CN 24.72 m: P, 3°, RF, PR, CN 24.81 m: P, 5°, RF, ST, CN 24.86 m: HB 25.00 m: HB
												25.47 m: J, 10°, SM, CU, CT CL 25.59 m: P, 0°, SM, PR, CN 25.61-25.62 m: HB 25.62-26.20 m: J, 80-90°, SM, ST, healed
												26.00 m: HB
	18		Is(50) D=0.94 MPa Is(50) A=0.83 MPa	27.40			LAMINITE: SANDSTONE (70%): fine to medium grained, pale grey, interlaminated at 0-10° with SHALE (30%): dark grey.		100 (99) [99]			26.90-26.94 m: DB 26.92 m: J, 55°, SM, PR, CN 27.00-27.84 m: HB, x 5
												28.29-28.30 m: CS, 10°, RF, CT CL, 10 mm aperture
												28.57 m: P, 5°, RF, CT CL 28.59-28.60 m: P, 7°, RF, CL, 10 mm aperture
	19		Is(50) A=1.42 MPa Is(50) D=0.74 MPa	28.90			SANDSTONE: medium grained, pale grey and grey, very thinly cross-bedded at 5-20°, occasional carbonaceous laminations.		100 (100) [100]			28.87-29.01 m: DB
												29.33 m: P, 10°, RF, CN
												30.00 m: HB
	20		Is(50) A=0.74 MPa Is(50) D=0.77 MPa	30.40			from 29.48 to 30.90 m: indistinct bedding		100 (86) [86]			30.40 m: P, 0°, RF, CN
												30.62 m: P, 12°, RF, CT CL 30.71-30.92 m: P, 7-15°, RF, PR, CN, x 3
												30.97-31.05 m: DB
			Is(50) A=1.35 MPa Is(50) D=0.94 MPa	31.90				88			31.90 m: DB	

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Evans Lane, St. Leonards, 2065

Project No: 60764076
Logged by: V.Tsambalas
Start Date: 07/11/25
Checked by: J.Riley
End Date: 11/11/25

Driller: Terratest
Drill Rig: Comaccio Geo 205
Total Depth: 40.12 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 332911.30
Northing: 6256230.30
Hor. Datum: MGA2020-56
RL: 67.824 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) RQD (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ Soil: VL 2, L 4, M 20, H 60, VH 200, EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	21	Not Observed	Is(50) A=1.39 MPa Is(50) D=1.02 MPa C: 32.54-33.00 m				SANDSTONE: medium grained, pale grey and grey, very thinly cross-bedded at 5-20°, occasional carbonaceous laminations. from 32.11 to 33.22 m: indistinct bedding	FR	(88) [88]			32.00-32.03 m: HB, x 2
				35.0							32.53 m: J, 30°, RF, CN	
	33.40		Is(50) A=0.79 MPa Is(50) D=0.68 MPa	33.0			NO CORE: 33.22 - 33.40 m.					33.00-33.22 m: HB, x 3
							SAND: fine to medium grained, pale grey.	XW	100 (87) [77]			
	22		Is(50) A=0.94 MPa Is(50) D=0.62 MPa	34.0			SANDSTONE: medium grained, pale grey, indistinct bedding, occasional carbonaceous flecks.					33.68 m: J, 18°, RF, CN 33.71 m: J, 5°, RF, CN 33.76 m: DB 33.80 m: J, 50°, RF, CN 34.00 m: HB 34.03 m: J, 0-5°, RF, CT CL, CU 34.32 m: J, 45°, RF, CN 34.36 m: J, 8°, RF, CN 34.43 m: J, 6°, RF, CN 34.47-34.48 m: SZ, 5-15°, RF, CN
				35.0					100 (93) [89]			34.90 m: DB 35.00 m: HB 35.04-35.10 m: CS, 5-20°, RF, CT SP-CL 35.11 m: DB
	23		Is(50) A=1.07 MPa Is(50) D=0.85 MPa	32.0				FR				35.88 m: J, 5°, RF, CN 35.91 m: EW, 15°, RF, CL, 2 mm aperture 36.00 m: HB 36.03-36.05 m: CS, 10-20°, RF, GP 36.15 m: J, 30°, RF, CN 36.21 m: J, 10°, RF, CN 36.29-36.37 m: DB 36.40 m: J, 0-5°, SM, CT CL
				36.0					100 (100) [100]			
	24		Is(50) A=0.94 MPa Is(50) D=0.94 MPa C: 36.65-37.00 m	31.0								37.00 m: HB 37.05 m: HB 37.09 m: J, 8°, RF, CN
				37.0								
	25		Is(50) A=1.03 MPa Is(50) D=0.73 MPa	30.0								37.90 m: DB 38.00 m: HB
				38.0			NO CORE: 38.13 - 38.18 m.		100 (100) [100]			38.13 m: P, 5°, SM, CT CL 38.18 m: P, 10°, SM, CN
26	C: 38.18-38.39 m Is(50) A=0.34 MPa Is(50) D=0.25 MPa	29.0				SILTSTONE: fine grained, dark grey, laminated at 5°.		100 (100) [100]				
		39.0				SANDSTONE: medium grained, pale grey, very thinly cross-bedded at 5-25°, occasional carbonaceous flecks.				39.00 m: HB		
27	Is(50) A=0.98 MPa Is(50) D=1.29 MPa						FR				39.40-39.44 m: DB	
								100 (100) [100]				

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water

Project: Willoughby Submain Amplification Project

Location: Evans Lane, St. Leonards, 2065

Project No: 60764076

Logged by: V.Tsambalas

Start Date: 07/11/25

Checked by: J.Riley

End Date: 11/11/25

Driller: Terratest

Drill Rig: Comaccio Geo 205

Total Depth: 40.12 m

Hole Diameter: 115-96 mm

Inclination: 90°

Bearing: N/A

Easting: 332911.30

Northing: 6256230.30

Hor. Datum: MGA2020-56

RL: 67.824 m

Ver. Datum: AHD

Surface: Pavement (Asphalt)

HQ3 METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: ● D: ○ UCS: □ 60 300	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
		Not Observed	Is(50) A=1.2 MPa Is(50) D=1.22 MPa				SANDSTONE: medium grained, pale grey, very thinly cross-bedded at 5-25°, occasional carbonaceous flecks. Terminated at 40.12 m.	FR		VL ₂ L ₆ M ₂₀ H ₆₀ VH ₃₀₀ EH	20 60 200 600 2000	40.00 m: HB
					27.0							
					41.0							
					26.0							
					42.0							
					25.0							
					43.0							
					24.0							
					44.0							
					23.0							
					45.0							
					22.0							
					46.0							
					21.0							
					47.0							
					20.0							
					19.0							

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
VE		ES: 0.10-0.20 m PID=2.1 PPM			0.0			FILL - ASPHALTIC CONCRETE: dark grey, aggregates up to 80 mm, 100 mm thick.	M		FILL
		FILL - Sandy GRAVEL: fine to coarse grained, sub-angular to angular, dark grey; Sand, fine to coarse grained; with clay.									
	U	ES: 0.50-0.60 m PID=1.0 PPM			-82.0			FILL - CLAY: medium plasticity, orange-brown; with sand, fine to medium grained; trace gravel, fine to medium, sub-rounded to sub-angular.	w > PL		
		D: 0.80-1.50 m FV=75 kPa ES: 1.00-1.10 m PID=1.3 PPM									
AD/T		ES: 1.40-1.50 m PID=1.9 PPM D: 1.50-3.00 m			-81.0		CH	CLAY: medium plasticity, grey mottled brown; with sand, fine to coarse grained.	w < PL	F	RESIDUAL SOIL
		ES: 2.90-3.00 m PID=0.2 PPM D: 3.00-4.50 m									
		ES: 3.90-4.00 m PID=0.5 PPM			-79.0				VSt		
Log continued on next page.											
					-78.0						
					-77.0						
					-76.0						

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOL ₍₆₄₎ VL ₍₂₎ L ₍₄₎ M ₍₂₎ H ₍₆₎ VH ₍₂₀₎ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					82.0							
					1.0							
					81.0							
					2.0							
					80.0							
					3.0							
					79.0							
					4.0							
					5.0							
					6.0							
					7.0							
					8.0							
					9.0							
					10.0							
					11.0							
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					61.0							
					62.0							
					63.0							
					64.0							
					65.0							
					66.0							
					67.0							
					68.0							
					69.0							
					70.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL, L, M, H, VH, EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)	
HQ3	9.25	3	Is(50) A=0.04 MPa Is(50) D=0.05 MPa	75.0		LAMINITE: SANDSTONE (60%), fine grained, pale grey-white, interlaminated at 30-45°, with SILTSTONE (40%), grey and dark grey.	HW					9.77 m: MB 10.00 m: MB 10.56 m: P, 20°, IR 12.00 m: MB 12.33 m: MB	
				9.0		Clayey SAND: fine to medium grained, pale grey and pale grey-white; Clay, bedded at 25-30° (Extremely Weathered Sandstone).	XW						
	11.00	4	Is(50) A=0.1 MPa Is(50) D=0.05 MPa	74.0		NO CORE: 9.26 to 9.77 m.		100 (47) [47]					
				10.0		SANDSTONE: fine to medium grained, pale grey-white, indistinctly bedded.	HW						
	12.00	4	Is(50) A=0.11 MPa Is(50) D=0.06 MPa	73.0		from 10.36 m: becoming thinly bedded at 10-30° Sandy CLAY: low plasticity, pale grey-white; Sand, fine to medium grained, rock fabric visible (Extremely Weathered Sandstone).	XW	100 (68) [68]					
				11.0		from 11.10 m: yellow-brown staining SANDSTONE: fine to medium grained, pale grey-white, indistinctly bedded.	HW - MW						
	15.00	5	Is(50) D=0.15 MPa	72.0				100 (11) [11]					
			Is(50) A=0.07 MPa			NO CORE: 12.33 - 14.22 m.							
	69.0	6			71.0								
				15.0		Sandy CLAY: low to medium plasticity, dark grey and pale grey-white; Sand, fine to medium grained, bedding at 10-40° (Extremely Weathered Laminite).	XW	100 (0) [0]					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH 2 4 6 8 10 200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	16.50				68.0		Sandy CLAY: low to medium plasticity, dark grey and pale grey-white; Sand, fine to medium grained, bedding at 10-40° (Extremely Weathered Laminite).	XW	100 (82) [82]			
	7		Is(50) D=0.63 MPa Is(50) A=1.35 MPa	67.0	17.0		SANDSTONE: fine to medium grained, pale grey with dark grey laminations, very thinly bedded at 5-25°.					HAWKESBURY SANDSTONE 16.87-18.75 m: MB, x 17
	18.00		Is(50) A=0.98 MPa Is(50) D=1 MPa	66.0	18.0			SW - FR	100 (100) [100]			18.94 m: P, 35°, CT CL 19.00-20.00 m: MB, x 7
	19.50		Is(50) A=0.76 MPa Is(50) D=0.75 MPa	65.0	19.0				65 (65) [58]			20.00-20.15 m: MB, x 2 20.22-20.33 m: J, 60°, VR, CT CL 20.42-20.48 m: MB, x 2
	9		Is(50) D=0.97 MPa Is(50) A=1.07 MPa	64.0	20.0		NO CORE: 20.48 - 21.42 m.		86 (86) [63]			21.42 m: MB 21.51 m: J, 5°, RF, CT Qz, 3 mm aperture 21.56 m: J, 10°, PO, CT Qz 21.68-21.95 m: J, 5-30°, RF, CT CL, 5 mm aperture, up to 80 mm spacing, x 5 22.00-23.15 m: MB, x 5
	21.00		Is(50) D=0.7 MPa Is(50) A=0.67 MPa	63.0	21.0							23.14 m: J, 15°, RF, CT CL, 2 mm aperture 23.32 m: J, 15°, RF, CT CL, 3 mm aperture 23.54-23.79 m: MB
	10		Is(50) A=1.4 MPa Is(50) D=1.13 MPa	62.0	22.0		SANDSTONE: medium grained, grey, occasional carbonaceous laminations at 0-20°.	SW - FR				
			Is(50) D=1.18 MPa Is(50) A=1.47 MPa	62.0	23.0							
				62.0	24.0							
				62.0	24.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25

Checked by: J Riley
End Date: 20/11/25

Driller: Terratest	Hole Diameter: 125-300 mm	Easting: 333353.80	RL: 82.37 m
Drill Rig: Comacchio GEO 305	Inclination: 60°	Northing: 6256065.50	Ver. Datum: AHD
Total Depth: 70 m	Bearing: 270°	Hor. Datum: MGA2020-56	Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ SOIL 0-4 VL 2 L 6 M 20 W 60 CH 100 FH 200	DEFECT SPACING (mm) 20 60 200 600 2000	ADDITIONAL OBSERVATIONS (Defect Descriptions)					
HQ3	24.00	11	Is(50) A=0.87 MPa Is(50) D=0.82 MPa	-61.0	25.0		SANDSTONE: medium grained, grey, occasional carbonaceous laminations at 0-20° .	SW - FR	100 (98) [93]			24.00-25.00 m: MB, x 4					
	27.00											Is(50) A=0.7 MPa Is(50) D=0.55 MPa	-60.0	26.0	SW	100 (95) [90]	25.41 m: MB
																	25.61 m: MB
																	25.69 m: P, 15°, RF, VN CL, 2 mm aperture
																	26.15-26.17 m: EW, 15°, CL
																	26.20 m: J, 28°, VR, CT CL, 2 mm aperture
																	26.28 m: MB
																	26.32-26.35 m: EW, 5°, CL
																	26.58 m: MB
																	27.00-27.30 m: MB, x 6
27.30 m: MB																	
27.55 m: MB																	
27.66-27.71 m: EW, 10°, CL																	
27.81-28.00 m: MB, x 3																	
30.00	12	Is(50) A=0.82 MPa Is(50) D=0.59 MPa	-59.0	27.0	28.0	MW	95 (83) [83]	28.14 m: J, 10°, RF, CT CL, 1 mm aperture									
								28.67 m: MB									
								29.00 m: MB									
								29.22 m: MB									
								29.67 m: MB									
								29.71-29.80 m: SS, 15°, CT CL									
								29.92-30.00 m: MB									
								30.05 m: MB									
								30.26 m: MB									
								30.29-30.37 m: EW, 20°, CL									
30.48 m: MB																	
30.58-30.98 m: MB, x 9																	
13		Is(50) D=0.38 MPa Is(50) A=0.31 MPa	-57.0	31.0		NO CORE: 31.90 - 32.00 m.	HW - MW				31.06-31.11 m: EW, 15°, CL						
											31.26-31.35 m: EW, 27°, CL						
											31.48 m: MB						
			Is(50) D=0.26 MPa Is(50) A=0.26 MPa Is(50) D=0.26 MPa	-55.0							31.64 m: MB						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	32.00						SANDSTONE: medium grained, pale grey, occasional dark grey carbonaceous laminations 0-40°.		100 (100) [100]			32.10-32.43 m: MB, x 4
	14		Is(50) A=1.14 MPa Is(50) D=1.26 MPa C: 32.60-32.80 m	54.0				SW - FR				32.87 m: MB
	33.00				33.0				97 (90) [87]			33.35 m: MB 33.40 m: MB 33.62 m: MB
			Is(50) A=1.6 MPa Is(50) D=0.54 MPa				NO CORE: 33.69 - 33.72 m. SANDSTONE: medium grained, pale grey, occasional dark grey carbonaceous laminations 0-40°.					33.82 m: J, 70°, RF, CT Fe, 1 mm aperture, 0.2 m persistence 33.90-34.06 m: MB, x 3
	15				34.0			SW - FR				34.24 m: P, 10°, RF, CT Ca, 1 mm aperture 34.47 m: P, 10°, RF, VN Ca, 3 mm aperture
			Is(50) A=0.98 MPa Is(50) D=0.93 MPa									34.75-35.00 m: MB, x 3
			Is(50) A=1.41 MPa Is(50) D=0.89 MPa		35.0							35.05 m: J, 30°, VR, VN Ca, 2 mm aperture 35.20-35.60 m: P, 5-25°, RF, CT CL, up to 4 mm mm aperture, 10 -120 mm spacing, x 7
	36.00				36.0		NO CORE: 35.83 - 35.90 m. SANDSTONE: medium grained, pale grey, occasional dark grey carbonaceous laminations 0-40°.	SW - FR	89 (80) [50]			35.72 m: J, 80°, IR, CT CL, 0.2 m persistence
			Is(50) A=1.31 MPa Is(50) D=0.68 MPa		51.0		NO CORE: 36.04 - 36.22 m. SANDSTONE: medium grained, pale grey, occasional dark grey carbonaceous laminations 0-40°.					36.32 m: MB 36.38-36.54 m: SS, 0-15°, RF, VN CL 36.56-36.82 m: J, 0-30°, RF, VN CL, 10 - 40 mm spacing, x 6
	16				37.0							37.00-38.20 m: MB, x 9
17	37.70				50.0				100 (100) [85]			
			Is(50) A=0.65 MPa Is(50) D=0.73 MPa					SW - FR				38.64-38.96 m: P, 0-15°, RF, CT CL, 20 - 50 mm spacing, x 5 38.70 m: MB
			Is(50) A=0.62 MPa Is(50) D=1.06 MPa		49.0							39.00-39.05 m: MB, x 2 39.08 m: P, 10°, VN CL, 2 mm 39.13 m: P, 10°, RF, VN CL, 6 mm aperture 39.24 m: MB
			Is(50) A=1.04 MPa Is(50) D=1.29 MPa		48.0							39.63-40.00 m: MB, x 5
					40.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • IS ₍₅₀₎ D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	40.00		Is(50) A=0.86 MPa Is(50) D=0.98 MPa				SANDSTONE: fine to medium grained, pale grey, trace carbonaceous inclusions, up to 3 mm.	SW - FR	100 (98) [93]			40.00 m: MB
	18		Is(50) A=1 MPa Is(50) D=1.09 MPa	47.0	41.0							40.70 m: J, 26°, RF, CT dark grey, 20 mm aperture
			Is(50) A=1.2 MPa Is(50) D=0.54 MPa C: 42.57-42.80 m	46.0	42.0		LAMINITE: SILTSTONE (80%), fine grained, dark grey, interlaminated at 0-30° with SANDSTONE (20%), fine to medium grained, pale grey and grey-brown.	MW - SW	100 (100) [97]			41.00 m: MB
	43.00		Is(50) A=0.88 MPa Is(50) D=0.34 MPa	43.0	45.0							42.00-42.07 m: CS, GP, medium to coarse grained, angular
19			Is(50) A=0.88 MPa Is(50) D=0.81 MPa	44.0	44.0			SW - FR	100 (99) [98]			42.19 m: P, 10°, RF, VN CL, 10 mm aperture
			Is(50) A=0.36 MPa Is(50) D=0.53 MPa	45.0	45.0		SANDSTONE: fine to medium grained, pale grey, very thinly bedded to thinly laminated at 0-30°, trace carbonaceous and quartz inclusions.					42.28 m: MB
			Is(50) A=0.88 MPa Is(50) D=0.58 MPa	43.0	46.0							42.38 m: MB
20	46.00		Is(50) A=0.68 MPa Is(50) D=0.85 MPa	46.0	47.0							42.89-43.24 m: MB, x 5
				47.0	47.0							43.53 m: P, 10°, RF, CT Qz
				47.0	47.0							43.58 m: P, 5°, SL, VN CL
				47.0	47.0							43.80 m: P, 15°, RF, VN CL, 5 mm aperture
				47.0	47.0							43.95 m: P, 10°, RF, VN CL, 8 mm aperture
				47.0	47.0							44.00 m: MB
				47.0	47.0							44.34 m: MB
				47.0	47.0							44.51 m: P, 8°, RF, CT Qz, 2 mm aperture
				47.0	47.0							44.63 m: P, 12°, RF, CT Qz, 6 mm aperture
				47.0	47.0							44.85 m: MB
				47.0	47.0							45.00 m: MB
				47.0	47.0							
				47.0	47.0							46.00 m: MB
				47.0	47.0							46.13 m: P, 15°, PR
				47.0	47.0							10% water loss
				47.0	47.0							46.20 m: MB
				47.0	47.0							46.28 m: MB
				47.0	47.0							46.34 m: MB
				47.0	47.0							46.50 m: P, 5°, RF, VN CL, 4 mm aperture
				47.0	47.0							46.70-47.00 m: MB, x 4
				47.0	47.0							46.74 m: MB
				47.0	47.0							
				47.0	47.0							47.43-47.46 m: EW, 5°, RF, VN CL
				47.0	47.0							47.71 m: MB

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25

Checked by: J Riley
End Date: 20/11/25

Driller: Terratest	Hole Diameter: 125-300 mm	Easting: 333353.80	RL: 82.37 m
Drill Rig: Comacchio GEO 305	Inclination: 60°	Northing: 6256065.50	Ver. Datum: AHD
Total Depth: 70 m	Bearing: 270°	Hor. Datum: MGA2020-56	Surface: Pavement (Asphalt)

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
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Driller: Terratest
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Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: * D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	23		Is(50) A=1.2 MPa Is(50) D=0.86 MPa	57.0	33.0		SANDSTONE: fine to medium grained, pale grey, very thinly bedded to thinly laminated at 0-30°, trace carbonaceous and quartz inclusions.	SW - FR				56.00 m: MB 56.14 m: P, 10°, RF, CT Ca, 2 mm aperture 56.48-57.10 m: P, 5-25°, RF, VN Ca, up to 6 mm mm aperture, 20 - 300 mm spacing, x 6
	58.00		PP=200, 300, 180 kPa Is(50) A=0.99 MPa Is(50) D=1.59 MPa	58.0	32.0		Gravelly CLAY: low plasticity, dark grey; Gravel, fine to medium; (Extremely Weathered Sandstone). SANDSTONE: medium grained, thinly to very thinly bedded at 5-15°.	XW	100 (100) [100]			57.74 m: MB 57.85-57.88 m: J, 15-25°, RF, CT Ca, 3 mm aperture, x 2 58.05 m: J, 40°, PR 58.26 m: J, 40°, PR 58.58 m: J, 40°, CU 59.00 m: MB 59.30-59.45 m: MB
	59.45		Is(50) A=1.7 MPa Is(50) D=1.34 MPa	59.0	31.0				100 (97) [97]			60.00 m: MB 60.16 m: J, 10°, RF, CT CL, 20 mm aperture 60.40-60.44 m: SS, GP, medium to coarse grained, angular 60.53 m: J, 15°, RF, CT Q, 3 mm aperture 60.75 m: MB 60.96 m: P, 5°, RF, VN CL, 5 mm aperture 61.00 m: MB 61.15 m: P, 5°, PR, VN CL, 3 mm
	61.00		C: 60.00-60.42 m Is(50) A=1.45 MPa Is(50) D=1.25 MPa	60.0	30.0		SANDSTONE: fine to medium grained, pale grey, occasional dark grey carbonaceous laminations 0-15°, trace quartz inclusions.	SW - FR	100 (100) [92]			61.87 m: MB 61.90 m: P, 10°, RF, CT Qz, 10 mm aperture 62.00-62.64 m: MB, x 3
26	61.00		Is(50) A=1.33 MPa Is(50) D=0.07 MPa	29.0	61.0		from 61.15 - 61.22 m: cross-bedding visible					62.84-63.28 m: J, 0-30°, RF, CT Qz, up to 3 mm mm aperture, 50-150 mm spacing, x 7
	63.0		Is(50) A=0.6 MPa Is(50) D=1.17 MPa	28.0	63.0		from 62.35 - 63.22 m: cross-bedding visible					63.68 m: J, 10°, RF, CT Qz, 30 mm aperture
			Is(50) A=1.76 MPa Is(50) D=0.99 MPa	27.0	64.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Atchison Lane and Oxley Street, Crows Nest, 2065

Project No: 60764076
Logged by: J Looney
Start Date: 10/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 305
Total Depth: 70 m
Hole Diameter: 125-300 mm
Inclination: 60°
Bearing: 270°
Easting: 333353.80
Northing: 6256065.50
Hor. Datum: MGA2020-56
RL: 82.37 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ J: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 M: 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 H: 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 VH: 61 62 63 64 65 66 67 68 69 70 EH: 71 72 73 74 75 76 77 78 79 80	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	64.00						SANDSTONE: fine to medium grained, pale grey, occasional dark grey carbonaceous laminations 0-15°, trace quartz inclusions. NO CORE: 64.09 - 64.59 m.	FR	73 (5) [0]			
	27		Is(50) D=0.01 MPa		65.0		CLAY: low to medium plasticity, dark grey, (Extremely Weathered Siltstone).	XW				
	65.85		Is(50) A=0.03 MPa Is(50) D=0.04 MPa		66.0		SANDSTONE: fine to medium grained, pale grey-white, indistinct bedding, trace carbonaceous inclusions.		100 (81) [76]			66.02 m: MB 66.07 m: J, 15°, PR, VN CL 66.13 m: J, 15°, PR, VN CL 66.26-66.31 m: EW, 5°, RF, VN CL 66.38 m: P, 15°, RF, VN CL, 4 mm aperture
	28		Is(50) A=0.22 MPa Is(50) D=1.01 MPa		67.0				100 (100) [97]			67.00 m: MB
29	67.00		C: 67.00-67.20 m		67.0							
			Is(50) A=1.01 MPa Is(50) D=1.28 MPa		68.0			SW - FR				68.00 m: MB 68.15 m: J, 5°, RF, PR, CT CL, 3 mm aperture 68.20 m: MB 68.23 m: MB
			Is(50) A=1.49 MPa Is(50) D=0.59 MPa		69.0							68.51-68.59 m: J, 0-15°, RF, CT CL, up to 2 mm mm aperture, 70 mm spacing, x 2 68.73-68.80 m: MB, x 3 68.90 m: J, 5°, RF, CT CL, 10 mm aperture 69.00-69.04 m: MB, x 2
			Is(50) A=0.85 MPa Is(50) D=0.89 MPa		70.0		Terminated at 70.00 m. Target depth.					69.78 m: J, 15°, RF, VN CL, 25 mm aperture

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water

Project: Willoughby Submain Amplification Project

Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076

Logged by: R Rees

Start Date: 24/11/25

Checked by: J Riley

End Date: 26/11/25

Driller: Terratest

Hole Diameter: 96-120 mm

Easting: 333461.00

RL: 76.9 m

Drill Rig: Commachio GEO 305

Inclination: 90°

Northing: 6256039.30

Ver. Datum: AHD

Total Depth: 55.11 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement (Asphalt)

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
C	C				0.0			FILL - ASPHALTIC CONCRETE: dark grey, aggregates up to 10mm, 80mm thick.			FILL
	VE	ES: 0.10-0.20 m PID=0.1 PPM						FILL - Sandy GRAVEL: fine to coarse grained, angular to sub-angular, dark grey, igneous; Sand, fine to coarse grained; trace silt, with cobbles.	M		
		ES: 0.50-0.60 m PID=0.1 PPM						FILL - Silty SAND: fine to coarse grained, angular to sub-angular, brown; trace gravel.			
		ES: 1.00-1.10 m PID=0.1 PPM		-76.0	1.0			FILL - CLAY: high plasticity, brown; with sand; trace gravel.	w ≈ PL		
	HWT	SPT: 1.50-1.95 m 2.4,6 N=10 ES: 1.50-1.60 m PID=0.1 PPM		-75.0	2.0			CLAY: high plasticity, pale grey mottled red-brown; with gravel; trace sand.	w > PL		RESIDUAL SOIL
		ES: 2.00-2.10 m PID=0.1 PPM									
		ES: 3.00-3.10 m PID=0.1 PPM		-74.0	3.0	CH				F - St	
	AD/T	ES: 4.00-4.10 m PID=0.1 PPM		-73.0	4.0	CH		Gravelly CLAY: high plasticity, pale grey mottled red-brown; with sand.	w ≈ PL		
		SPT: 4.50-4.57 m 18/70 mm HB N=R		-72.0	5.0	CL-Cl		Sandy CLAY: low to medium plasticity; Sand, fine to coarse grained; trace gravel, (Sandstone).		H	EXTREMELY WEATHERED MATERIAL
								Log continued on next page.			
				-71.0	6.0						
				-70.0	7.0						
				-69.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: R Rees
Start Date: 24/11/25
Checked by: J Riley
End Date: 26/11/25

Driller: Terratest
Drill Rig: Commachio GEO 305
Total Depth: 55.11 m
Hole Diameter: 96-120 mm
Inclination: 90°
Bearing: N/A
Easting: 333461.00
Northing: 6256039.30
Hor. Datum: MGA2020-56
RL: 76.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					76.0							
					75.0							
					74.0							
					73.0							
					72.0							
					5.0							
HQ3	1	6.28	25/11/2025 07:30:09				Log continued from previous page. Sandy CLAY: low to medium plasticity; Sand, fine to coarse grained; trace gravel. SANDSTONE: fine to medium grained, pale grey, thinly bedded at 0-25°. from 5.48 m: becoming dark red-brown, iron stained from 5.63 m: becoming red mottled pale grey	HW HW - MW	100 (92) [92]			HAWKESBURY SANDSTONE 5.20-5.27 m: DB, x 3 5.32 m: P, 15°, RF, CN 5.35-6.28 m: HB, x 9
	2							MW	100 (100) [100]			6.58 m: P, 15°, SM, CT clay
												7.00-7.58 m: HB, x 5
												7.62 m: P, 5°, clay, 10 mm aperture

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: R Rees
Start Date: 24/11/25
Checked by: J Riley
End Date: 26/11/25

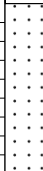
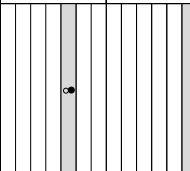
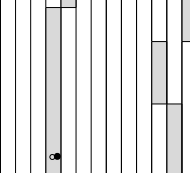
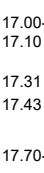
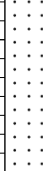
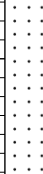
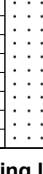
Driller: Terratest
Drill Rig: Commachio GEO 305
Total Depth: 55.11 m
Hole Diameter: 96-120 mm
Inclination: 90°
Bearing: N/A
Easting: 333461.00
Northing: 6256039.30
Hor. Datum: MGA2020-56
RL: 76.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: ○ UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	9.28		Is(50) A=0.68 MPa Is(50) D=0.55 MPa	68.0	9.0		SANDSTONE: fine to medium grained, red-brown and grey, thinly laminated to laminated at 0-15° . from 8.14 m: becoming pale grey, thinly laminated	MW	100 (100) [100]			7.95-8.00 m: HB 8.12-8.13 m: EW, 0°, clay 8.53 m: HB 8.87 m: P, 10°, SM, CT manganese
			Is(50) A=0.23 MPa Is(50) D=0.68 MPa	67.0	10.0		from 9.28 m: becoming pale grey mottled yellow-brown, fine to coarse grained, thinly bedded to thickly laminated at 0-20°, trace black flecks, iron stained					9.28 m: MB 9.44 m: P, 15°, RF, SN, Fe 9.51-10.02 m: MB, x 4 9.62 m: MB
	12.28		Is(50) A=0.73 MPa Is(50) D=0.72 MPa	66.0	11.0		from 11.23 to 11.53 m: becoming thickly laminated at 0-15°	HW	100 (99) [99]			10.00 m: HB 10.02 m: HB 10.06 m: P, 5°, RF, CN 10.08-12.00 m: HB, x 8
			C: 11.30-11.50 m Is(50) A=1.46 MPa Is(50) D=0.85 MPa	65.0	12.0		SILTSTONE: fine grained, grey and dark grey, thinly laminated to very thinly bedded at 0-5° . SANDSTONE: fine to medium grained, pale grey, trace black flecks, thickly to thinly laminated at 0-20° .					12.05 m: J, 65°, SM, SN Fe 12.28 m: MB
4	15.28		Is(50) A=1.17 MPa Is(50) D=1.33 MPa	64.0	13.0		from 13.03 m: occasional iron staining	SW - FR	100 (100) [100]			12.95 m: P, 5°, clay, 3 mm aperture 13.00-13.38 m: MB, x 4
			Is(50) A=1.23 MPa Is(50) D=1.25 MPa	63.0	14.0							14.32 m: P, 15°, RF, CT ML
			Is(50) A=1.79 MPa Is(50) D=1.25 MPa	62.0	15.0							14.76-14.78 m: EW 15.00-15.28 m: HB, x 3
			Is(50) A=1.59 MPa Is(50) D=1.07 MPa	61.0	16.0							15.52-16.60 m: MB, x 7

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: R Rees
Start Date: 24/11/25
Checked by: J Riley
End Date: 26/11/25
Driller: Terratest
Hole Diameter: 96-120 mm
Easting: 333461.00
RL: 76.9 m
Drill Rig: Commachio GEO 305
Inclination: 90°
Northing: 6256039.30
Ver. Datum: AHD
Total Depth: 55.11 m
Bearing: N/A
Hor. Datum: MGA2020-56
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: ● B: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)																																																																																																																																																																																																																																																																																																																																																																																																																											
HQ3	5		Is(50) A=2.38 MPa Is(50) D=1.52 MPa	60.0	17.0		SANDSTONE: fine to medium grained, pale grey, trace black flecks, thickly to thinly laminated at 0-20° .	SW - FR	100 (99) [99]			17.00-17.04 m: HB, x 2 17.10 m: P, 10°, RF, CT ML																																																																																																																																																																																																																																																																																																																																																																																																																											
													from 17.0 to 17.33 m: becoming thickly to thinly laminated at 0-10°	17.31 m: MB 17.43 m: J, 55°, RF, CN																																																																																																																																																																																																																																																																																																																																																																																																																									
															from 17.33 m: very thinly bedded to thinly laminated at 0-20°	17.70-18.35 m: MB, x 7																																																																																																																																																																																																																																																																																																																																																																																																																							
																	18.36-18.38 m: SZ 18.68 m: P, 20°, RF, CN, 2 mm aperture 18.97-19.00 m: HB, x 2 19.32 m: P, 0°, IR, clay, 3 mm aperture 19.86-21.28 m: HB, x 8																																																																																																																																																																																																																																																																																																																																																																																																																						
	6		Is(50) A=0.82 MPa Is(50) D=0.54 MPa	59.0	18.0		from 21.0 m: becoming thinly bedded to thickly laminated		100 (96) [95]			21.43 m: P, 0-5°, IR, clay, 3 mm aperture 21.55-21.68 m: SZ 21.89-22.00 m: HB, x 2																																																																																																																																																																																																																																																																																																																																																																																																																											
													Is(50) A=1.09 MPa Is(50) D=0.48 MPa	58.0	19.0			Is(50) A=1.21 MPa Is(50) D=1.45 MPa C: 19.65-19.85 m	57.0	20.0		Is(50) A=1.05 MPa Is(50) D=1.02 MPa	56.0	21.0		Is(50) A=0.99 MPa Is(50) D=1 MPa	55.0	22.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0	23.0		Is(50) A=1.75 MPa Is(50) D=1.62 MPa	53.0	24.0		Is(50) A=1.57 MPa Is(50) D=1.49 MPa	54.0

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: R Rees
Start Date: 24/11/25
Checked by: J Riley
End Date: 26/11/25

Driller: Terratest
Drill Rig: Commachio GEO 305
Total Depth: 55.11 m
Hole Diameter: 96-120 mm
Inclination: 90°
Bearing: N/A
Easting: 333461.00
Northing: 6256039.30
Hor. Datum: MGA2020-56
RL: 76.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	24.32		Is(50) A=1.47 MPa Is(50) D=1.26 MPa				SANDSTONE: fine to medium grained, pale grey, trace black flecks, thinly bedded to thickly laminated at 0-10° .	SW - FR	100 (100) [89]			24.00-26.00 m: HB, x 6
							from 25.0 m: becoming fine to coarse grained, pale grey, trace black and pale grey-white flecks, very thinly bedded at 0-15° flecks					
	26.43		Is(50) A=1.35 MPa Is(50) D=1.2 MPa				LAMINITE: Interbedded SILTSTONE (70%) and SANDSTONE (30%); siltstone is fine grained, dark grey; sandstone is pale grey, fine to medium grained; thinly laminated at 0-10° .	HW	100 (98) [0]			26.20 m: P, 5°, SM
												26.43-26.61 m: DB, x 3
	27.30		C: 26.61-26.80 m Is(50) A=0.05 MPa Is(50) D=0.03 MPa					SW - FR	100 (100) [69]			26.84-26.86 m: CS, 10°, SM
												27.00-27.88 m: HB, x 6
			Is(50) A=0.31 MPa Is(50) D=0.3 MPa					SW - FR				27.93 m: J, 70°, SM, CN
												27.95-28.39 m: HB, x 5
			Is(50) A=0.05 MPa Is(50) D=0.04 MPa				SANDSTONE: fine to medium grained, pale grey and grey, very thinly bedded to thinly laminated at 0-20° .	SW - FR				28.59 m: P, 10°, SM, CT ML
												28.63-29.02 m: MB, x 3
	30.29		Is(50) A=0.86 MPa Is(50) D=0.75 MPa					SW - FR				29.41 m: P, 15°, SM, CN
												29.67-31.00 m: HB, x 7
11			Is(50) A=0.95 MPa Is(50) D=0.6 MPa				from 31.04 - 31.07 m: carbonaceous laminations	SW - FR	100 (95) [89]			31.00-31.07 m: EW
												31.42-31.46 m: EW
			Is(50) A=1 MPa Is(50) D=0.68 MPa					SW - FR				31.68 m: P, 5°, RF, carbonaceous , 3 mm aperture
												31.74-31.79 m: EW
								SW - FR				31.80-32.00 m: MB

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Project: Willoughby Submain Amplification Project
Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: R Rees
Start Date: 24/11/25

Checked by: J Riley
End Date: 26/11/25

Driller: Terratest
Drill Rig: Commachio GEO 305
Total Depth: 55.11 m

Hole Diameter: 96-120 mm
Inclination: 90°
Bearing: N/A

Easting: 333461.00
Northing: 6256039.30
Hor. Datum: MGA2020-56

RL: 76.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: 0-4 VL 2 L 4 M 20 H 60 VH 200 EH 600 2000	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	12		Is(50) A=0.23 MPa Is(50) D=0.42 MPa	44.0	33.0	[.....]	SANDSTONE: fine to medium grained, pale grey and grey, very thinly bedded to thinly laminated at 0-20° . from 32.12 to 32.20 m: with carbonaceous flecks	SW - FR	100 (97) [93]	[.....]	[.....]	32.26-32.37 m: P, 0-10°, 20-40 mm spacing, x 5
			Is(50) A=0.71 MPa Is(50) D=0.75 MPa	43.0	34.0		from 33.0 m: becoming very thinly bedded to thinly laminated at 0-25°	HW - MW				32.58 m: MB 32.65-32.70 m: J, 55°, RF, CN, 50 - 100 mm spacing, x 4 32.72 m: P, 5° 32.78-33.28 m: MB, x 6
	13		Is(50) A=0.72 MPa Is(50) D=0.63 MPa Is(50) A=1.45 MPa Is(50) D=1.08 MPa	42.0	35.0	[.....]	from 35.80 m: occasional carbonaceous laminations	SW - FR	100 (71) [67]	[.....]	[.....]	33.37 m: J, 65°, RF, CN 33.42-33.46 m: EW 33.53-33.56 m: EW 33.78 m: J, 80°, RF, CN, 0.2 m persistence 33.96-34.16 m: HB, x 3 34.15 m: P, 15°, IR 34.23-34.26 m: EW 34.37-34.84 m: MB, x 4
			Is(50) A=0.12 MPa Is(50) D=0.16 MPa	41.0	36.0		from 37.14 m: becoming thickly to thinly bedded	HW - MW				34.92 m: P, 10°, SM, CT ML, 10 mm spacing, x 2 35.00 m: HB 35.28 m: MB 35.75-36.28 m: MB, x 6
	14		Is(50) A=0.72 MPa Is(50) D=0.65 MPa	40.0	37.0	[.....]	from 37.95 to 38.04 m: dark grey	HW	100 (100) [95]	[.....]	[.....]	36.35 m: P, 15°, SM, CN 36.80-37.00 m: MB, x 4 37.00 m: HB 37.05 m: J, 90°, RF, CN, 0.1 m persistence 37.10-37.16 m: SZ 37.16-37.55 m: MB, x 4
			Is(50) A=1.73 MPa Is(50) D=0.41 MPa C: 39.60-39.80 m	39.0	38.0		from 38.90 - 39.12 m: becomes thinly laminated at 0-5° from 39.13 m: trace carbonaceous flecks and laminations	HW - MW				37.90 m: P, 5° 38.18-38.88 m: SZ
				38.0	39.0	[.....]	at 39.53: siltstone inclusion, fine grained, dark grey, 50 mm	FR				38.88 m: MB 38.95-39.08 m: SZ
				37.0	40.0							39.28 m: MB 39.46 m: P, 5° 39.58 m: P, 0°, SM, CT ML

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: R Rees
Start Date: 24/11/25
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Hole Diameter: 96-120 mm
Inclination: 90°
Bearing: N/A
Easting: 333461.00
Northing: 6256039.30
Hor. Datum: MGA2020-56
RL: 76.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: ○ UCS: □ Is(50) 0-200 A: 0-60 D: 0-200 UCS: 0-200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	15		Is(50) A=1.27 MPa Is(50) D=0.88 MPa				SANDSTONE: fine to medium grained, pale grey and grey, thinly to thickly bedded at 0-25° . from 40.50 - 40.60 m: trace siltstone inclusions, <50 mm. from 40.69 - 40.70 m: siltstone inclusion, <10 mm from 40.70 m: no carbonaceous flecks or laminations					40.00-40.04 m: HB, x 3 40.38 m: P, 15°, RF, carbonaceous, 1-3 mm aperture 40.62 m: MB 41.00 m: HB 41.20 m: P, 15°, SM, CT ML
	42.30		Is(50) A=0.96 MPa Is(50) D=0.87 MPa				from 41.90 m: indistinct bedding					41.88 m: J, 5°, SM, CN 42.00 m: HB 42.03 m: J, 15°, RF, CT ML, SN, Fe
			Is(50) A=0.94 MPa Is(50) D=0.91 MPa				from 42.35 - 43.80 m: trace iron staining		100 (99) [99]			42.30 m: MB 43.00 m: HB
HQ3	16		Is(50) A=1.47 MPa Is(50) D=1.18 MPa									44.00 m: HB
			Is(50) A=1.19 MPa Is(50) D=0.84 MPa									44.30 m: J, 20°, RF, ML, 2 mm aperture 44.32 m: MB 44.43 m: MB 44.45 m: J, 20°, RF, CN 44.57 m: P, 5°, SM, clay, 5 mm aperture 44.73-44.76 m: EW, 10°, clay
	45.26		Is(50) A=0.34 MPa Is(50) D=0.76 MPa				from 45.0 m: becoming thickly bedded at 0-15°		100 (100) [76]			45.00-45.28 m: HB, x 6 45.40 m: J, 20°, RF, CN 45.63 m: J, 55°, RF, CN 45.78 m: P, 0°, CT clay
HQ3	17		C: 46.00-46.18 m				from 46.0 m: becoming very thinly bedded to thickly laminated					46.00-46.33 m: HB, x 4
			Is(50) A=2.55 MPa Is(50) D=1.92 MPa				from 46.5 m: trace carbonaceous flecks					46.53 m: P, 15°, quartz vein , 3 mm aperture 46.72 m: MB
			Is(50) A=1.6 MPa Is(50) D=1.38 MPa									46.88 m: P, 10°, RF, CN 47.00 m: MB
												47.35-48.07 m: P, 10-15°, RF, VN ML, 20 - 100 mm spacing, x 8

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Hume Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: R Rees
Start Date: 24/11/25
Checked by: J Riley
End Date: 26/11/25
Easting: 333461.00
Northing: 6256039.30
Hor. Datum: MGA2020-56
RL: 76.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

Driller: Terratest
Drill Rig: Commachio GEO 305
Total Depth: 55.11 m
Hole Diameter: 96-120 mm
Inclination: 90°
Bearing: N/A

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	48.30						SANDSTONE: fine to medium grained, pale grey and grey, very thinly bedded to thickly laminated at 0-15°, trace carbonaceous flecks.		100 (100) [88]			48.00-48.30 m: HB, x 3
			Is(50) A=0.47 MPa Is(50) D=0.78 MPa	28.0	49.0							48.55-49.00 m: P, 0-10°, RF, CN, 50 - 100 mm spacing, x 6
	18		Is(50) A=0.83 MPa Is(50) D=0.85 MPa	27.0	50.0		from 49.55 m: becoming thickly laminated to thinly laminated					49.37 m: P, 10°, RF, CT ML, 1 mm aperture
			Is(50) A=0.69 MPa Is(50) D=0.99 MPa	26.0	51.0		from 51.0 m: occasional iron staining					49.63-51.00 m: MB, x 6
	51.30		Is(50) A=0.99 MPa Is(50) D=1.07 MPa	25.0	52.0			FR	100 (100) [100]			50.15 m: P
	52.10		Is(50) A=0.8 MPa Is(50) D=1.05 MPa	24.0	53.0				100 (100) [100]			51.00 m: HB 51.05 m: P, 10°, RF, CN 51.30-53.00 m: MB, x 6
	52.88		Is(50) A=1.37 MPa Is(50) D=0.34 MPa	23.0	54.0		from 53.77 m: becoming fine to coarse grained, thinly bedded at 0-15°, trace carbonaceous flecks, trace rounded siltstone inclusions < 35 mm		100 (100) [100]			53.54 m: MB
	21		Is(50) A=1.47 MPa Is(50) D=0.99 MPa	22.0	55.0							53.77 m: P, 0° 53.85-55.00 m: MB, x 5
			Is(50) A=2.19 MPa Is(50) D=1.28 MPa	21.0	56.0		Terminated at 55.11 m. Target depth.					55.06 m: P, 0°

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
VE	HWT	ES: 0.20-0.30 m PID=0.1 PPM	Not Observed	80.0	1.0		CH	ASPHALTIC CONCRETE: dark grey, aggregates up to 20 mm, 100 mm thick.	w PL	n	FILL
		ES: 0.50-0.60 m PID=0.1 PPM						FILL - CLAY: low plasticity, pale grey, mottled red-brown; with gravel, medium to coarse, sub-angular to sub-rounded.			RESIDUAL SOIL
		ES: 1.00-1.10 m PID=1 PPM						CLAY: medium plasticity, pale grey mottled red-brown.			
								Log continued on next page.			
				79.0	2.0						
				78.0	3.0						
				77.0	4.0						
				76.0	5.0						
				75.0	6.0						
				74.0	7.0						
				73.0	8.0						

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					80.0							
					1.0							
							<i>Log continued from previous page.</i>					
					79.0		SANDSTONE: medium to coarse grained, dark red-brown occasionally pale grey, bedded at 0-15°, iron stained.		100 (100) [91]			HAWKESBURY SANDSTONE 1.67 m: DB 1.78-1.85 m: P, 0°, SM, PR, CN, 30 mm spacing, x 3 1.86 m: P, 0°, SM, PR, CN 1.90 m: DB 2.15 m: P, 15°, RF, PR, CN 2.24 m: MB 2.39 m: P, 15°, RF, PR, CN 2.52 m: J, 5°, RF, IR, CT Qz 2.61 m: P, 0°, RF, IR, CT Qz 2.73-2.00 m: MB, x 3
					2.0							
					78.0		from 3.00 m: becoming pale grey	HW - MW				3.06 m: P, 0°, IR 3.25-4.00 m: MB, x 4
					3.0				100 (100) [99]			
							from 3.48 m: becoming pale grey mottled pale red-brown (iron stained)					
					77.0							
					4.0							
							from 4.60 m: no longer iron stained pale red-brown					4.65 m: P, 0°, SM, PR, CT CL, 10 mm 4.83 m: P, 0°, PR 5.00-6.00 m: MB, x 3
					76.0							
					5.0							
					75.0							
					6.0							
							from 6.40 m: occasional shale laminations, 2-3 mm thick	MW - SW				6.26 m: P, 0°, CT CL, 2 mm 6.28 m: P, 0°, CT CL, 8 mm 6.35-6.40 m: MB, x 2 6.47 m: P, 0°, SM, PR, CT CL, 3 mm
					74.0				100 (100) [100]			
					7.0							7.00-8.00 m: MB, x 4
					73.0							
					8.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • Is(50) D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HC3	4	Not Observed	Is(50) A=0.8 MPa Is(50) D=0.4 MPa	72.0	9.0		SANDSTONE: medium to coarse grained, pale grey occasionally dark grey, bedded at 0-15°, occasional shale inclusions 2-3 mm thick, occasional cross-bedding.	MW - SW				8.17 m: MB
			Is(50) A=0.57 MPa Is(50) D=0.54 MPa	71.0	10.0		from 9.00 m: occasional dark brown staining		100 (99) [99]			8.36 m: P, 0°, SM, IR, CT CL, 3 mm
			Is(50) A=1.6 MPa Is(50) D=1.28 MPa	70.0	11.0			SW				9.00-10.00 m: MB, x 5
			Is(50) A=0.69 MPa Is(50) D=0.87 MPa	69.0	12.0							10.26-10.27 m: EW, 0°, CT CL
			Is(50) A=0.58 MPa Is(50) D=0.54 MPa	68.0	13.0		from 13.05 to 13.90 m: cross-bedded at 0-15°		100 (99) [99]			10.41-10.42 m: EW, 0°, CT CL
			Is(50) A=0.96 MPa Is(50) D=0.76 MPa	67.0	14.0							10.72 m: MB
			Is(50) A=1.22 MPa Is(50) D=0.84 MPa	66.0	15.0		from 14.75 to 14.81 m: trace carbonaceous inclusions	FR				11.82-11.83 m: EW, 0°, SM, PR, CT CL
			Is(50) A=1.22 MPa Is(50) D=0.86 MPa	65.0					100 (100) [100]			12.00-12.45 m: MB, x 2
												12.51-12.53 m: EW, 5°, SM, PR, CT CL
												12.72 m: MB
												13.00 m: MB
												13.90 m: P, 0°, SM, PR, CN
												14.00 m: MB
												14.78 m: P, 0°, CT
												15.00 m: MB
												15.50 m: MB
												15.61 m: P, 10°, PR
												15.68 m: DB
												15.72 m: P, 15°, SM, PR, CT CL

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) RQD (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ S ₆₀ 200 L ₂ 60 M ₂ 20 H ₂ 60 VH 200 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)						
HQ3	6	Not Observed	Is(50) A=0.75 MPa Is(50) D=0.68 MPa	64.0	17.0		SANDSTONE: medium to coarse grained, dark red-brown occasionally pale grey, bedded at 0-15°, iron-stained.	FR	100 (100) [99]			16.00-16.29 m: MB, x 2						
			Is(50) A=0.41 MPa Is(50) D=0.25 MPa	63.0	18.0		16.50 m: P, 5°, SM, PR, CT CL											
	Is(50) A=1.24 MPa Is(50) D=0.97 MPa		62.0	19.0	from 18.55 m: occasional iron staining		17.00 m: MB 17.11 m: P, 0°, SM, PR, CT CL											
	Is(50) A=1.32 MPa Is(50) D=1.21 MPa		61.0	20.0	18.00-21.00 m: MB, x 11													
	Is(50) A=1.45 MPa Is(50) D=1.15 MPa		60.0	21.0	21.02 m: P, 0°, SM, PR, VN CL 21.05 m: P, 0°, SM, PR, VN CL													
	Is(50) A=1.21 MPa Is(50) D=0.9 MPa C: 21.75-22.00 m		59.0	22.0	from 22.13 m: becoming red-brown and yellow brown, iron stained		21.48 m: P, 0°, SM, PR, CN 21.55-23.00 m: MB, x 3											
	Is(50) A=1.07 MPa Is(50) D=1.1 MPa		58.0	23.0	from 23.52 to 24.1 m: becoming red-brown and yellow brown, iron stained; occasional quartz inclusions up to 10 mm thick		23.61 m: P, 0-5°, RF, IR, VN Qz											
	Is(50) A=1.08 MPa Is(50) D=0.68 MPa		57.0															
	8																	

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • IS ₍₅₀₎ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	24.45	Not Observed	Is(50) A=1.02 MPa Is(50) D=0.84 MPa				SANDSTONE: medium to coarse grained, dark red-brown occasionally pale grey, bedded at 0-15°, iron-stained.		100 (100) [100]			24.00-26.00 m: MB, x 8
			Is(50) A=1.41 MPa Is(50) D=0.94 MPa Is(50) A=0.94 MPa									
			Is(50) A=0.67 MPa Is(50) D=0.61 MPa				SHALE: dark grey, thinly laminated at 0°, occasional SANDSTONE laminations, pale grey, fine to medium grained.					26.55 m: P, 0°, SM, PR, VN CL 26.65-29.00 m: MB, x 12
	27.45		Is(50) A=0.54 MPa Is(50) D=0.48 MPa						100 (100) [100]			
10		Not Observed	Is(50) A=0.23 MPa Is(50) D=0.71 MPa					FR				
			C: 29.00-29.25 m									
			Is(50) A=0.92 MPa Is(50) D=0.35 MPa									29.53 m: P, 5°, SM, IR, CN 29.56-30.00 m: MB, x 3
	30.50		Is(50) A=1.26 MPa Is(50) D=1 MPa				SANDSTONE: medium to coarse grained, pale grey, bedded at 0-20°.		100 (100) [100]			30.07 m: P, 0°, SM, PR, CN 30.21 m: P, 5°, SM, PR, CN 30.40-30.50 m: MB, x 2 30.65 m: J, 45°, SM, PR, CT CL 30.75 m: P, 5°, SM, PR, CN 30.80-32.00 m: MB, x 3
			Is(50)									

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: ● D: ○ UCS: □ SOL ₆₄ VL ₂ L ₄ M ₂₀ H ₆₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)			
HQ3	12	Not Observed	A=0.86 MPa Is(50) D=1.74 MPa				SANDSTONE: medium to coarse grained, pale grey, bedded at 0-20°.	FR	100 (100) [92]			32.00-33.50 m: MB, x 8			
			Is(50) A=0.81 MPa Is(50) D=0.97 MPa	48.0	33.0										
			Is(50) A=0.24 MPa Is(50) D=1.07 MPa	47.0	34.0								33.69 m: P, 15°, SM, PR, CT CL, 3 mm		
			Is(50) A=0.34 MPa Is(50) D=1.81 MPa	46.0	35.0					CONGLOMERATE: fine to coarse grained, pale grey speckled white and dark grey, quartz and shale clasts, up to 20 mm, rounded.					34.65 m: P, 5°, SM, PR, VN CL 34.74 m: P, 15°, SM, PR, CT CL, 3 mm 34.83 m: J, 10°, IR 34.94 m: J, 0-15°, IR 35.00 m: MB 35.14 m: J, 0°, SM, IR, CN 35.21 m: J, 0-10°, RF, IR, CT Qz 35.33 m: J, 0°, RF, IR, CT Qz 35.46 m: J, 10°, RF, IR, CT Qz
			Is(50) D=0.45 MPa	45.0	36.0					SANDSTONE: medium to coarse grained, pale grey, bedded at 0-20°.					35.67-36.50 m: MB, x 5
			Is(50) A=1.19 MPa Is(50) D=1.14 MPa	44.0	37.0					from 37.44 to 38.57 m: becoming iron stained				36.59 m: P, 10°, RF, PR, CT Qz 36.70 m: J, 5°, SM, PR, CT Qz 36.81 m: P, 15°, SM, PR, CT CL 36.97-37.00 m: MB, x 2	
	Is(50) A=1.21 MPa Is(50) D=0.96 MPa		43.0	38.0						37.43 m: J, 0°, IR, CT CL, 9 mm 37.53 m: J, 5° 37.73 m: P, 15° 38.00 m: MB					
	Is(50) A=0.5 MPa Is(50) D=1.16 MPa		42.0	39.0		CONGLOMERATE: fine to coarse grained, pale grey speckled white and dark grey, quartz and shale clasts, up to 10 mm, rounded.				38.39-38.42 m: EW, 0°, CL 38.57 m: J, 40°, PR, CT CL 38.81-39.00 m: MB, x 2 39.00-40.00 m: MB					
	Is(50) A=1.16 MPa Is(50) D=1.49 MPa		41.0			SANDSTONE: medium to coarse grained, pale grey, bedded at 0-15°, trace carbonaceous inclusions.	95 (95) [95]								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □ L ₂ M ₂ H ₂ V ₂ EH ₂	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	14	Not Observed	Is(50) A=1.81 MPa Is(50) D=1.79 MPa				SANDSTONE: medium to coarse grained, pale grey, bedded at 0-15°, trace carbonaceous inclusions.	FR				40.00-41.00 m: MB, x 3
			C: 41.20-41.48 m Is(50) A=1.85 MPa Is(50) D=1.43 MPa	40.0	41.0		from 41.00 m: becoming thickly laminated with siltstone at 10°					41.59 m: P, 10°, PR 41.89 m: P, 10°, PR 42.00 m: MB
HQ3	45.31	Not Observed	Is(50) A=1.95 MPa Is(50) D=1.5 MPa				NO CORE: 42.60 m - 42.90 m.					
			Is(50) A=1.74 MPa Is(50) D=1.57 MPa	38.0	43.0		SANDSTONE: medium to coarse grained, pale grey, bedded at 0-5°, with carbonaceous inclusions, dark grey.	FR				43.00-44.00 m: MB, x 8
			Is(50) A=1.55 MPa Is(50) D=1.16 MPa	37.0	44.0		from 43.54 to 43.76 m: carbonaceous flecks					44.05 m: P, 1-2°, RF, PR, CN 44.27-46.12 m: MB, x 10
			Is(50) A=2.8 MPa Is(50) D=1.22 MPa	36.0	45.0		from 44.27 m: becoming iron stained	FR				
			Is(50) A=0.94 MPa Is(50) D=0.78 MPa	45.0	46.0		at 44.84 m: carbon bond at 45.00 m: becoming fine and medium grained		100 (100) [100]			46.28 m: J, 10°, RF, IR, CT CL
			Is(50) A=0.91 MPa Is(50) D=1.07 MPa	35.0	47.0		at 45.71 m: becoming medium to coarse grained, grey and dark grey from 46.00 to 46.30 m: becoming fine to medium grained, grey and dark grey					47.00-48.00 m: MB, x 8
				34.0	48.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
Checked by: J Riley
End Date: 18/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 305
Total Depth: 63.65 m
Hole Diameter: 96-250 mm
Inclination: 90°
Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: ● D: ○ UCS: □ SOL ₂₄₅ VL ₂ L ₆ M ₂₅ H ₄₅ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)												
HQ3	48.50	Not Observed	Is(50) A=1.03 MPa Is(50) D=1.02 MPa	32.0	49.0		SANDSTONE: medium to coarse grained, pale grey, bedded at 0-5°, with carbonaceous inclusions, dark grey.	FR	100 (100) [100]			48.50-50.46 m: MB, x 11												
							Is(50) A=1.22 MPa Is(50) D=0.98 MPa						31.0	50.0	from 49.8 to 49.95 m: siltstone clast									
															Is(50) A=1.43 MPa Is(50) D=1.36 MPa		30.0	51.0	CONGLOMERATE: fine to coarse grained, pale grey speckled white and dark grey, quartz and shale clasts, up to 30 mm, rounded.					
																			Is(50) A=1.52 MPa Is(50) D=1.22 MPa		29.0	52.0	SANDSTONE: medium to coarse grained, pale grey, bedded at 0-15°, trace carbonaceous inclusions.	
																							Is(50) A=1.09 MPa Is(50) D=1.34 MPa C: 52.30-53.00 m	
	Is(50) A=1.55 MPa Is(50) D=1.49 MPa		27.0	54.0																				
					Is(50) A=1.37 MPa Is(50) D=1.13 MPa		26.0		55.0															
	Is(50) A=1.48 MPa Is(50) D=1.2 MPa		25.0			from 55.30 to 55.40 m: carbon bands, 1-2 mm thick																		

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Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Willoughby Lane and Atchison Street, Crows Nest, 2065

Project No: 60764076
Logged by: D Aouad
Start Date: 10/11/25
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Driller: Terratest
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Total Depth: 63.65 m
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Bearing: N/A
Easting: 333638.60
Northing: 6255984.80
Hor. Datum: MGA2020-56
RL: 80.9 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □ SOL ₆₄ VL ₂ L ₄ M ₂₀ H ₆₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	18						SANDSTONE: medium to coarse grained, pale grey, bedded at 0-15°, trace carbonaceous inclusions.					56.00-59.00 m: MB, x 9
			Is(50) A=1.17 MPa Is(50) D=1.2 MPa		24.0							
					57.0							
	57.57		Is(50) A=1.39 MPa Is(50) D=1.18 MPa		23.0		from 57.77 m: becoming dark grey		100 (100) [100]			
					58.0							
			Is(50) A=1.32 MPa Is(50) D=1.34 MPa		22.0		from 58.33 m: becoming pale grey					
					59.0							
	59.43		Is(50) A=1.9 MPa Is(50) D=1.65 MPa		21.0				100 (100) [100]			59.62 m: P, 5°, RF, PR, SN
		20			60.0							59.87-61.35 m: MB, x 10
	60.45		Is(50) A=1.63 MPa Is(50) D=1.3 MPa		20.0		from 60.60 m: becoming brown-grey		100 (100) [100]			
21					61.0		from 61.00 m: becoming to pale grey					
			Is(50) A=3.13 MPa Is(50) D=3.36 MPa		19.0		CONGLOMERATE: fine to coarse grained, brown-grey speckled white and dark grey, with quartz and shale clasts, up to 3 mm.					61.47 m: P, 2°, RF, PR, CN
					62.0		SANDSTONE: medium to coarse grained, pale brown-grey, bedded at 0-10°.					61.74-63.65 m: MB, x 6
			Is(50) A=1.7 MPa Is(50) D=0.75 MPa		18.0							
					63.0							
			Is(50) A=1.49 MPa Is(50) D=1.44 MPa		17.0		Terminated at 63.65 m. Target depth.					
					64.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 205
Total Depth: 75.17 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 334038.60
Northing: 6255782.40
Hor. Datum: MGA2020-56
RL: 91.28 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
VE	HWT	ES: 0.20-0.30 m PID=0.1 PPM ES: 0.50-0.60 m PID=0.1 PPM ES: 1.00-1.10 m PID=0.1 PPM SPT: 1.50-1.95 m 5.7, 11 N=18 ES: 1.50-1.60 m PID=0.1 PPM PP=550 kPa PP=320 kPa PP=580 kPa SPT: 2.30-2.30 m 20/0 mm HB N=R ES: 2.30-2.40 m PID=0.9 PPM	Not Observed	0.0 91.0 1.0 90.0 2.0 89.0		CL-CI CL	ASPHALTIC CONCRETE: dark grey, 50 mm thick. FILL - Sandy Cobbley GRAVEL: coarse grained, angular to sub-angular, dark grey, concrete, igneous; Sand, fine to coarse grained; Cobbles, grey and dark grey, angular, concrete. FILL - Gravelly SAND: fine to coarse grained, dark brown; Gravel, fine to coarse, angular, igneous; with clay. CLAY: medium plasticity, red-brown mottled yellow-brown. CLAY: medium plasticity, pale grey mottled dark red-brown and yellow-brown; trace gravel, fine, rounded to sub-rounded, ironstone; trace cobbles/boulders, rounded, sandstone, ironstone; iron stained staining.	w < PL w ≈ PL w < PL	F VSt	PAVEMENT RESIDUAL SOIL
Log continued on next page.										
				3.0 88.0 4.0 87.0 5.0 86.0 6.0 85.0 7.0 84.0 8.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
Checked by: J Riley
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Driller: Terratest
Drill Rig: Comacchio GEO 205
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Inclination: 90°
Bearing: N/A
Easting: 334038.60
Northing: 6255782.40
Hor. Datum: MGA2020-56
RL: 91.28 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: ○ UCS: □ Soil class	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					91.0							
					1.0							
					90.0							
					2.0							
					89.0							
							<i>Log continued from previous page.</i>					
	1		Is(50) A=0.32 MPa Is(50) D=0.21 MPa		3.0		INTERBEDDED AND MUDSTONE: INTERBEDDED SILTSTONE and SANDSTONE: SILTSTONE (60%) fine grained, pale grey and grey, laminated at 0-10°; SANDSTONE (40%), fine to medium grained, dark grey and red-brown, trace iron staining; MUDSTONE, dark grey.	SW	99 (83) [76]			MITTAGONG FORMATION 2.50 m: DB 2.55 m: DB 2.62 m: DB 2.70 m: P, 0°, SM, UN 2.78 m: P, 0°, SM, UN 2.90 m: P, 0°, SM, PR 3.05 m: P, 0°, SM, PR 3.21 m: P, 10°, SM, PR 3.25 m: P, 10°, SM, PR 3.43 m: P, 10°, SM, PR 3.54 m: P, 10°, SM, PR 3.58 m: J, 20°, SM, UN 3.68 m: J, 30°, SM, PR 3.70 m: HB 3.77 m: P, 5°, SM, PR 3.81 m: P, 5°, SM, PR 3.95 m: J, 20°, RF, UN 4.16 m: DB 4.25 m: P, 0°, SM, PR 4.36 m: P, 5°, SM, UN 4.46 m: P, 15°, SM, UN 4.64 m: P, 10°, SM, PR, clay 2mm 4.70 m: P, 10°, SM, PR, clay 2mm
	2		Is(50) A=0.25 MPa Is(50) D=0.14 MPa		4.0			SW to FR	100 (100) [100]			
			Is(50) A=0.68 MPa Is(50) D=0.03 MPa C: 4.65-4.91 m		87.0							
					5.0		SANDSTONE: fine to coarse grained, pale grey, bedded at 0-15°, with carbonaceous laminations.		96 (96) [83]			HAWKESBURY SANDSTONE 5.09 m: HB
					6.0		from 5.52 to 5.66 m: thinly laminated from 5.75 to 5.80 m: dark red-brown, iron stained from 5.82 to 6.13: becoming pale grey mottled grey and dark grey, iron stained					5.56 m: P, 5°, SM, PR, sand 1mm 5.65 m: P, 5°, SM, PR 5.82 m: J, 30°, RF, PR, carbon 1mm 5.95 m: J, 80°, RF, UN 6.17 m: J, 5°, RF, PR
					85.0		from 6.48 to 6.86 m: becoming pale grey mottled grey and dark grey, iron stained	FR	99 (99) [90]			purple staining 6.85 m: P, 25°, RF, PR 7.07 m: P, 5°, SM, UN, clay 1mm 7.53 m: P, 10°, RF, PR, clayey sand 2-4mm 7.79 m: P, 15°, SM, PR 7.85 m: P, 15°, SM, PR, carbon 1mm
	4		Is(50) A=0.67 MPa Is(50) D=0.38 MPa		84.0		from 7.79 to 7.91 m: becoming pale grey mottled grey					
					8.0							

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Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest	Hole Diameter: 115-96 mm	Easting: 334038.60	RL: 91.28 m
Drill Rig: Comacchio GEO 205	Inclination: 90°	Northing: 6255782.40	Ver. Datum: AHD
Total Depth: 75.17 m	Bearing: N/A	Hor. Datum: MGA2020-56	Surface: Pavement (Asphalt)

[illegible]

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
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Start Date: 14/11/25
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Driller: Terratest
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Total Depth: 75.17 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 334038.60
Northing: 6255782.40
Hor. Datum: MGA2020-56
RL: 91.28 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil 60, 200, 600, 2000 L M H VH EH VL 2 4 6 8 10 200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	10		Is(50) A=0.91 MPa Is(50) D=0.78 MPa	75.0			SANDSTONE: fine to coarse grained, pale grey, bedded at 0-15°, with carbonaceous laminations.					16.18 m: HB
												16.62 m: HB
	17.00			17.0					100 (100) [100]			16.95 m: DB
			Is(50) A=1.5 MPa Is(50) D=1.1 MPa	74.0								17.74 m: HB
												17.93 m: HB
				18.0								18.04 m: HB
	18.50		Is(50) A=0.98 MPa Is(50) D=0.86 MPa	73.0					100 (100) [100]			18.62 m: P, 15°, RF, UN
												18.75 m: P, 15°, RF, PR
				19.0								18.79 m: HB
												18.90 m: HB
HQ3	12	Not Observed	Is(50) A=1.4 MPa Is(50) D=1.1 MPa	72.0								19.04 m: HB
												19.28 m: HB
				20.0					97 (97) [97]			19.40 m: P, 15°, RF, PR
	20.00		Is(50) A=1.4 MPa Is(50) D=1.1 MPa	71.0			higher proportion of carbonaceous laminations from 20.51 to 20.58 m: becoming grey and dark grey, higher proportion of carbonaceous laminations, thinly laminated	FR				20.57 m: HB
												21.08 m: HB
				21.0								21.09 m: P, 20°, RF, PR
	21.50		Is(50) A=2 MPa Is(50) D=1.1 MPa	70.0					100 (100) [100]			22.00 m: HB
												22.25 m: P, 5°, RF, PR
				22.0								22.64 m: P, 30°, RF, PR
				69.0								22.87 m: P, 30°, RF, PR
HQ3	14		Is(50) A=1.4 MPa Is(50) D=1.1 MPa	23.0			from 23.00 to 23.61 m: bedding undulating		100 (100) [96]			23.56 m: P, 10°, RF, UN
	23.00		Is(50) A=1.9 MPa Is(50) D=1.8 MPa	68.0			from 23.79 to 23.90 m: cross-bedding visible, bedded at 45° and 0°.					23.87 m: HB

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 205
Total Depth: 75.17 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 334038.60
Northing: 6255782.40
Hor. Datum: MGA2020-56
RL: 91.28 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS(50) A: • D: o UCS: □ SOL ₆₄ VL ₂ L ₄ M ₂₀ H ₂₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	24.50	16	Is(50) A=1.6 MPa Is(50) D=1.4 MPa	67.0			SANDSTONE: fine to coarse grained, pale grey, bedded at 0-15°, with carbonaceous laminations.	FR	100 (100) [100]			23.92 m: HB 24.08 m: HB 24.22 m: HB 24.25 m: HB 24.45 m: HB 24.49 m: HB
				25.0								24.82 m: HB 25.10 m: HB
	26.00	17	Is(50) A=1.6 MPa Is(50) D=1.3 MPa	66.0					98 (98) [98]			25.50 m: P, 30°, RF, PR 25.62 m: P, 30°, RF, PR 25.66 m: J, 0°, RF, UN 25.74 m: P, 30°, RF, UN
				26.0								26.07 m: P, 5°, SM, PR
	27.50	18	Is(50) A=1.2 MPa Is(50) D=1.4 MPa	65.0					100 (100) [100]			26.50 m: P, 40°, SM, PR, clay 1mm 26.76 m: HB, carbon 1mm 26.86 m: HB 27.05 m: HB
				27.0								27.50 m: DL 27.61 m: HB 27.73 m: HB
	29.00	19	Is(50) A=1.7 MPa Is(50) D=1.2 MPa	28.0					100 (100) [100]			28.03 m: HB 28.30 m: HB 28.43 m: HB
				63.0								from 28.67 to 29.37 m: becoming grey and dark grey
	30.50	20	Is(50) A=0.94 MPa Is(50) D=0.96 MPa	29.0					100 (100) [100]			from 29.14 to 29.17 m: thinly laminated 29.17 m: P, 0°, SM, PR 29.35 m: HB
				62.0								from 30.0 to 30.81 m: indistinct bedding, trace dark grey carbonaceous flecks
			Is(50) A=1.4 MPa Is(50) D=1.2 MPa	30.0					100 (100) [100]			30.83 m: P, 5°, SM, PR 30.96 m: HB 31.09 m: HB
				61.0								31.40 m: HB 31.70 m: HB
			Is(50) A=1.4 MPa Is(50) D=1.2 MPa	31.0								
				60.0								
				32.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
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Driller: Terratest
Drill Rig: Comacchio GEO 205
Total Depth: 75.17 m
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Inclination: 90°
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Easting: 334038.60
Northing: 6255782.40
Hor. Datum: MGA2020-56
RL: 91.28 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOL ₆₄ VL ₂ L ₄ M ₂₀ H ₆₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	32.00	Not Observed	Is(50) A=0.51 MPa Is(50) D=0.73 MPa	59.0			SANDSTONE: fine to coarse grained, pale grey, bedded at 0-15°, with carbonaceous laminations. from 32.05 to 32.26 m: thinly laminated	FR	94 (91) [91]			32.12 m: P, 5°, RF, UN 32.23 m: HB 32.47 m: P, 10°, RF, PR, carbon 1mm 32.72 m: HB
	33.50		Is(50) A=1.7 MPa Is(50) D=0.89 MPa	58.0			from 33.0 to 33.02 m: siltstone bed		100 (100) [97]			33.64 m: HB 33.86 m: HB 33.87 m: P, 30°, SM, UN 33.92 m: HB 34.00-34.05 m: J, 45°, RF, UN
	35.00		C: 34.60-34.90 m Is(50) A=0.84 MPa Is(50) D=0.56 MPa	57.0			at 34.27 m: zone of bioturbation at 34.70 m: zone of bioturbation		100 (100) [100]			35.29 m: HB 35.37 m: HB 35.55 m: DL 35.57 m: DB
	36.50		Is(50) A=1.1 MPa Is(50) D=1.1 MPa Is(50) A=1.9 MPa Is(50) D=0.96 MPa	56.0			SANDSTONE: medium to coarse grained, grey, bedded at 0-20°. from 35.4 to 35.58 m: siltstone rip up clasts, dark grey		99 (73) [73]			36.10 m: HB 36.25-38.80 m: J, 90°, RF, UN 36.66-36.90 m: J, 80°, RF, UN 36.86 m: P, 20°, SM, PR 36.88 m: HB
	38.00		Is(50) A=1.2 MPa Is(50) D=1.2 MPa Is(50) A=0.49 MPa Is(50) D=1 MPa	54.0			from 36.64 to 36.67: thinly laminated, dark grey from 37.12 to 37.32 m: cross-bedding visible at 37.56 m: rounded clast, dark grey, 40 mm		100 (65) [65]			37.75-37.85 m: P, 45°, RF, UN, carbon 1mm 10% water loss
	39.50		Is(50) A=1.4 MPa Is(50) D=1.5 MPa	52.0			from 39.6 m: bedded at 0-10 °		100 (27) [27]			38.90 m: HB 38.90-40.16 m: J, 70°, RF, UN

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
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Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	26				51.0		SANDSTONE: medium to coarse grained, grey, thickly to thinly laminated at 0-10°, trace carbonaceous clasts.					40.15 m: HB 40.25 m: HB 40.44 m: P, 5°, RF, UN
	41.00				41.0				100 (100) [100]			40.86 m: P, 5°, RF, PR 40.86-40.90 m: J, 75°, RF, UN
	27		Is(50) A=1.3 MPa Is(50) D=1.6 MPa		50.0							41.28 m: HB
					42.0							
	42.50		Is(50) A=1.5 MPa Is(50) D=1.4 MPa		49.0				99 (86) [79]			42.50 m: DL
					43.0							42.90 m: HB 42.95 m: P, 5°, RF, UN
	28		Is(50) A=1.6 MPa Is(50) D=1.4 MPa		48.0							43.28-43.40 m: J, 45°, RF, UN
	44.00	Not Observed			44.0		from 44.0 m: becoming thinly bedded	FR	100 (100) [100]			44.23 m: J, 5°, RF, UN
	29		Is(50) A=2.1 MPa Is(50) D=2 MPa		47.0							44.62 m: HB 44.88 m: HB, carbon 1mm 45.07 m: HB
	45.50		Is(50) A=2.1 MPa Is(50) D=1.7 MPa		46.0				100 (100) [100]			
30			Is(50) A=0.66 MPa Is(50) D=1.1 MPa		45.0		from 46.6 m: becoming thickly to thinly laminated					46.06 m: HB 46.24 m: HB
					46.0							46.62 m: HB
	47.00		Is(50) A=1.6 MPa Is(50) D=0.36 MPa		47.0		from 47.05 to 47.14 m: dark grey		96 (96) [96]			46.84 m: P, 30°, SM, PR
31					48.0							

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest

Hole Diameter: 115-96 mm

Easting: 334038.60

RL: 91.28 m

Drill Rig: Comacchio GEO 205

Inclination: 90°

Northing: 6255782.40

Ver. Datum: AHD

Total Depth: 75.17 m

Bearing: N/A

Hor. Datum: MGA2020-56

Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	48.50	32	Is(50) A=0.8 MPa Is(50) D=0.95 MPa C: 48.64-48.84 m	43.0			SANDSTONE: medium to coarse grained, grey, thickly to thinly laminated at 0-10°, trace carbonaceous clasts.	FR	100 (100) [97]			

Client: Sydney Water
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Project No: 60764076
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Driller: Terratest
Drill Rig: Comacchio GEO 205
Total Depth: 75.17 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 334038.60
Northing: 6255782.40
Hor. Datum: MGA2020-56
RL: 91.28 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ SOLIDITY: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	56.00		A=1.4 MPa Is(50) D=0.93 MPa	35.0			SANDSTONE: medium to coarse grained, grey, thickly to thinly laminated at 0-10°, trace carbonaceous clasts. from 59.55 to 62.40 m: becoming indistinctly bedded	FR	100 (100) [100]			55.95 m: HB
	37		Is(50) A=1.4 MPa Is(50) D=1.2 MPa	57.0								57.16 m: HB
	57.50		Is(50) A=1.9 MPa Is(50) D=1.2 MPa	34.0					100 (100) [100]			57.52 m: P, 30°, RF, UN
	38		Is(50) A=1.6 MPa Is(50) D=1.2 MPa	58.0								58.14 m: HB 58.21 m: HB
	59.00			33.0								58.52 m: HB 58.77 m: HB
				59.0					100 (100) [100]			
	39	Not Observed	Is(50) A=1.9 MPa Is(50) D=1.3 MPa	32.0								59.36 m: P, 5°, RF, UN 59.52 m: HB
	60.50		Is(50) A=1.1 MPa Is(50) D=1.1 MPa	60.0					100 (100) [100]			
	40			31.0								
	61.00			61.0					100 (100) [100]			
	41			30.0								
	62.00		Is(50) A=1.1 MPa Is(50) D=1.2 MPa	62.0					100 (100) [100]			61.87 m: J, 15°, RF, PR 62.21 m: HB
	42		Is(50) A=1.3 MPa Is(50) D=1.2 MPa Is(50) A=0.88 MPa Is(50) D=1.2 MPa C: 63.14-63.54 m	29.0								63.13 m: HB
	63.50			28.0					100 (100) [100]			
				64.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25

Checked by: J Riley
End Date: 20/11/25

Driller: Terratest	Hole Diameter: 115-96 mm	Easting: 334038.60	RL: 91.28 m
Drill Rig: Comacchio GEO 205	Inclination: 90°	Northing: 6255782.40	Ver. Datum: AHD
Total Depth: 75.17 m	Bearing: N/A	Hor. Datum: MGA2020-56	Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)						
HQ3	43	Not Observed	Is(50) A=1.5 MPa Is(50) D=1.3 MPa Is(50) A=1.7 MPa Is(50) D=1.4 MPa Is(50) A=1.5 MPa Is(50) D=0.89 MPa Is(50) A=1.6 MPa Is(50) D=1.3 MPa Is(50) A=1.6 MPa Is(50) D=1.4 MPa Is(50) A=1.3 MPa Is(50) D=1.5 MPa Is(50) A=1.7 MPa Is(50) D=1.4 MPa	27.0		SANDSTONE: medium to coarse grained, grey, thickly to thinly laminated at 0-10°, trace carbonaceous clasts.	FR	100 (100) [100]			63.95 m: HB 64.07 m: HB 64.19 m: HB 64.37 m: HB 64.72 m: HB 64.92 m: HB 66.06 m: P, 15°, RF, UN 66.10 m: HB 66.52 m: P, 15°, RF, UN 66.79 m: HB 66.97 m: P, 15°, SM, PR 67.08 m: HB 67.65 m: HB 69.07 m: HB, carbon 1mm 69.36 m: HB 69.48 m: HB 69.81 m: HB 69.83 m: HB 69.84 m: HB 70.21 m: HB 70.33 m: HB 70.96 m: HB 71.04 m: DL 71.70 m: HB							
	65.05			65.0		66.0					66.0	66.0	66.0	66.0	66.0	66.0	66.0	
	44			26.0		25.0					24.0	23.0	22.0	21.0	20.0	19.0	18.0	17.0
	66.50			26.0		25.0					24.0	23.0	22.0	21.0	20.0	19.0	18.0	17.0
	45			24.0		23.0					22.0	21.0	20.0	19.0	18.0	17.0	16.0	15.0
	68.00			23.0		22.0					21.0	20.0	19.0	18.0	17.0	16.0	15.0	14.0
	46			23.0		22.0					21.0	20.0	19.0	18.0	17.0	16.0	15.0	14.0
	69.23			22.0		21.0					20.0	19.0	18.0	17.0	16.0	15.0	14.0	13.0
	47			22.0		21.0					20.0	19.0	18.0	17.0	16.0	15.0	14.0	13.0
	69.58			21.0		20.0					19.0	18.0	17.0	16.0	15.0	14.0	13.0	12.0
48	21.0	20.0	19.0	18.0	17.0	16.0	15.0	14.0	13.0	12.0								
71.04	20.0	19.0	18.0	17.0	16.0	15.0	14.0	13.0	12.0	11.0								
49	20.0	19.0	18.0	17.0	16.0	15.0	14.0	13.0	12.0	11.0								
71.50	19.0	18.0	17.0	16.0	15.0	14.0	13.0	12.0	11.0	10.0								
			Is(50) A=2.3 MPa Is(50)	19.0	18.0	17.0	16.0	15.0	14.0	13.0	12.0	11.0						
				18.0	17.0	16.0	15.0	14.0	13.0	12.0	11.0	10.0						
				17.0	16.0	15.0	14.0	13.0	12.0	11.0	10.0	9.0						
				16.0	15.0	14.0	13.0	12.0	11.0	10.0	9.0	8.0						
				15.0	14.0	13.0	12.0	11.0	10.0	9.0	8.0	7.0						
				14.0	13.0	12.0	11.0	10.0	9.0	8.0	7.0	6.0						
				13.0	12.0	11.0	10.0	9.0	8.0	7.0	6.0	5.0						
				12.0	11.0	10.0	9.0	8.0	7.0	6.0	5.0	4.0						
				11.0	10.0	9.0	8.0	7.0	6.0	5.0	4.0	3.0						
				10.0	9.0	8.0	7.0	6.0	5.0	4.0	3.0	2.0						
				9.0	8.0	7.0	6.0	5.0	4.0	3.0	2.0	1.0						
				8.0	7.0	6.0	5.0	4.0	3.0	2.0	1.0	0.0						
				7.0	6.0	5.0	4.0	3.0	2.0	1.0	0.0	-1.0						
				6.0	5.0	4.0	3.0	2.0	1.0	0.0	-1.0	-2.0						
				5.0	4.0	3.0	2.0	1.0	0.0	-1.0	-2.0	-3.0						
				4.0	3.0	2.0	1.0	0.0	-1.0	-2.0	-3.0	-4.0						
				3.0	2.0	1.0	0.0	-1.0	-2.0	-3.0	-4.0	-5.0						
				2.0	1.0	0.0	-1.0	-2.0	-3.0	-4.0	-5.0	-6.0						
				1.0	0.0	-1.0	-2.0	-3.0	-4.0	-5.0	-6.0	-7.0						
				0.0	-1.0	-2.0	-3.0	-4.0	-5.0	-6.0	-7.0	-8.0						
				-1.0	-2.0	-3.0	-4.0	-5.0	-6.0	-7.0	-8.0	-9.0						
				-2.0	-3.0	-4.0	-5.0	-6.0	-7.0	-8.0	-9.0	-10.0						
				-3.0	-4.0	-5.0	-6.0	-7.0	-8.0	-9.0	-10.0	-11.0						
				-4.0	-5.0	-6.0	-7.0	-8.0	-9.0	-10.0	-11.0	-12.0						
				-5.0	-6.0	-7.0	-8.0	-9.0	-10.0	-11.0	-12.0	-13.0						
				-6.0	-7.0	-8.0	-9.0	-10.0	-11.0	-12.0	-13.0	-14.0						
				-7.0	-8.0	-9.0	-10.0	-11.0	-12.0	-13.0	-14.0	-15.0						
				-8.0	-9.0	-10.0	-11.0	-12.0	-13.0	-14.0	-15.0	-16.0						
				-9.0	-10.0	-11.0	-12.0	-13.0	-14.0	-15.0	-16.0	-17.0						
				-10.0	-11.0	-12.0	-13.0	-14.0	-15.0	-16.0	-17.0	-18.0						
				-11.0	-12.0	-13.0	-14.0	-15.0	-16.0	-17.0	-18.0	-19.0						
				-12.0	-13.0	-14.0	-15.0	-16.0	-17.0	-18.0	-19.0	-20.0						
				-13.0	-14.0	-15.0	-16.0	-17.0	-18.0	-19.0	-20.0	-21.0						
				-14.0	-15.0	-16.0	-17.0	-18.0	-19.0	-20.0	-21.0	-22.0						
				-15.0	-16.0	-17.0	-18.0	-19.0	-20.0	-21.0	-22.0	-23.0						
				-16.0	-17.0	-18.0	-19.0	-20.0	-21.0	-22.0	-23.0	-24.0						
				-17.0	-18.0	-19.0	-20.0	-21.0	-22.0	-23.0	-24.0	-25.0						
				-18.0	-19.0	-20.0	-21.0	-22.0	-23.0	-24.0	-25.0	-26.0						
				-19.0	-20.0	-21.0	-22.0	-23.0	-24.0	-25.0	-26.0	-27.0						
				-20.0	-21.0	-22.0	-23.0	-24.0	-25.0	-26.0	-27.0	-28.0						
				-21.0	-22.0	-23.0	-24.0	-25.0	-26.0	-27.0	-28.0	-29.0						
				-22.0	-23.0	-24.0	-25.0	-26.0	-27.0	-28.0	-29.0	-30.0						
				-23.0	-24.0	-25.0	-26.0	-27.0	-28.0	-29.0	-30.0	-31.0						
				-24.0	-25.0	-26.0	-27.0	-28.0	-29.0	-30.0	-31.0	-32.0						
				-25.0	-26.0	-27.0	-28.0	-29.0	-30.0	-31.0	-32.0	-33.0						
				-26.0	-27.0	-28.0	-29.0	-30.0	-31.0	-32.0	-33.0	-34.0						
				-27.0	-28.0	-29.0	-30.0	-31.0	-32.0	-33.0	-34.0	-35.0						
				-28.0	-29.0	-30.0	-31.0	-32.0	-33.0	-34.0	-35.0	-36.0						
				-29.0	-30.0	-31.0	-32.0	-33.0	-34.0	-35.0	-36.0	-37.0						
				-30.0	-31.0	-32.0	-33.0	-34.0	-35.0	-36.0	-37.0	-38.0						
				-31.0	-32.0	-33.0	-34.0	-35.0	-36.0	-37.0	-38.0	-39.0						
				-32.0	-33.0	-34.0	-35.0	-36.0	-37.0	-38.0	-39.0	-40.0						
				-33.0	-34.0	-35.0	-36.0	-37.0	-38.0	-39.0	-40.0	-41.0						
				-34.0	-35.0	-36.0	-37.0	-38.0	-39.0	-40.0	-41.0	-42.0						
				-35.0	-36.0	-37.0	-38.0	-39.0	-40.0	-41.0	-42.0	-43.0						
				-36.0	-37.0	-38.0	-39.0	-40.0	-41.0	-42.0	-43.0	-44.0						
				-37.0	-38.0	-39.0	-40.0	-41.0	-42.0	-43.0	-44.0	-45.0						
				-38.0	-39.0	-40.0	-41.0	-42.0	-43.0	-44.0	-45.0	-46.0						
				-39.0	-40.0	-41.0	-42.0	-43.0	-44.0	-45.0	-46.0	-47.0						
				-40.0	-41.0	-42.0	-43.0	-44.0	-45.0	-46.0	-47.0	-48.0						
				-41.0	-42.0	-43.0	-44.0	-45.0	-46.0	-47.0	-48.0	-49.0						
				-42.0	-43.0	-44.0	-45.0	-46.0	-47.0	-48.0	-49.0	-50.0						
				-43.0	-44.0	-45.0	-46.0	-47.0	-48.0	-49.0	-50.0	-51.0						
				-44.0	-45.0	-46.0	-47.0	-48.0	-49.0	-50.0	-51.0	-52.0						
				-45.0	-46.0	-47.0	-48.0	-49.0	-50.0	-51.0	-52.0	-53.0						
				-46.0	-47.0	-48.0	-49.0	-50.0	-51.0	-52.0	-53.0	-54.0						
				-47.0	-48.0	-49.0	-50.0	-51.0	-52.0	-53.0	-54.0	-55.0						
				-48.0	-49.0	-50.0	-51.0	-52.0	-53.0	-54.0	-55.0	-56.0						
				-49.0	-50.0	-51.0	-52.0	-53.0	-54.0	-55.0	-56.0	-57.0						
				-50.0	-51.0	-52.0	-53.0	-54.0	-55.0	-56.0	-57.0	-58.0						
				-51.0	-52.0	-53.0	-54.0	-55.0	-56.0	-57.0	-58.0	-59.0						
				-52.0	-53.0	-54.0	-55.0	-56.0	-57.0	-58.0	-59.0	-60.0						
				-53.0	-54.0	-55.0	-56.0	-57.0	-58.0	-59.0	-60.0	-61.0						
				-54.0	-55.0	-56.0	-57.0	-58.0	-59.0	-60.0	-61.0	-62.0						
				-55.0	-56.0	-57.0	-58.0	-59.0	-60.0	-61.0	-62.0	-63.0						
				-56.0	-57.0	-58.0	-59.0	-60.0	-61.0	-62.0	-63.0	-64.0						
				-57.0	-58.0	-59.0	-60.0	-61.0	-62.0	-63.0	-64.0	-65.0						
				-58.0	-59.0	-60.0	-61.0	-62.0	-63.0	-64.0	-65.0	-66.0						
				-59.0	-60.0	-61.0	-62.0	-63.0	-64.0	-65.0	-66.0	-67.0						
				-60.0	-61.0	-62.0	-63.0	-64.0	-65.0	-66.0	-67.0	-68.0						
				-61.0	-62.0	-63.0	-64.0	-65.0	-66.0	-67.0	-68.0	-69.0						
				-62.0	-63.0	-64.0	-65.0	-66.0	-67.0	-68.0	-69.0	-70.0						
				-63.0	-64.0	-65.0	-66.0	-67.0	-68.0	-69.0	-70.0	-71.0						
				-64.0	-65.0	-66.0	-67.0	-68.0	-69.0	-70.0	-71.0	-72.0						
				-65.0	-66.0	-67.0	-68.0	-69.0	-70.0	-71.0	-72.0	-73.0						
				-66.0	-67.0	-68.0	-69.0	-70.0	-71.0	-72.0	-73.0	-74.0						
				-67.0	-68.0	-69.0	-70.0	-71.0	-72.0	-73.0	-74.0	-75.0						
				-68.0	-69.0	-70.0	-71.0	-72.0	-73.0	-74.0	-75.0	-76.0						
				-69.0	-70.0	-71.0	-72.0	-73.0	-74.0	-75.0	-76.0	-77.0						
				-70.0	-71.0	-72.0	-73.0	-74.0	-75.0	-76.0	-77.0	-78.0						
				-71.0	-72.0	-73.0	-74.0	-75.0	-76.0	-77.0	-78.0	-79.0						
				-72.0	-73.0	-74.0	-75.0	-76.0	-77.0	-78.0	-79.0	-80.0						
				-73.0	-74.0	-75.0	-76.0	-77.0	-78.0	-79.0	-80.0	-81.0						
				-74.0	-75.0	-76.0	-77.0	-78.0	-79.0	-80.0	-81.0	-82.0						
				-75.0	-76.0	-77.0	-78.0	-79.0	-80.0	-81.0	-82.0	-83.0						
				-76.0	-77.0	-78.0	-79.0	-80.0	-81.0	-82.0	-83.0	-84.0						
				-77.0	-78.0	-79.0	-80.0	-81.0	-82.0	-83.0	-84.0	-85.0						
				-78.0	-79.0	-80.0	-81.0	-82.0	-83.0	-84.0	-85.0	-86.0						
				-79.0	-80.0	-81.0	-82.0	-83.0	-84.0	-85.0	-86.0	-87.0						
				-80.0	-81.0	-82.0	-83.0	-84.0	-85.0	-86.0	-87.0	-88.0						
				-81.0	-82.0	-83.0	-8											

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: Corner of Thomas Lane and Huntington Street, Crows Nest, 2065

Project No: 60764076
Logged by: M. O'Neill
Start Date: 14/11/25
Checked by: J Riley
End Date: 20/11/25

Driller: Terratest
Drill Rig: Comacchio GEO 205
Total Depth: 75.17 m
Hole Diameter: 115-96 mm
Inclination: 90°
Bearing: N/A
Easting: 334038.60
Northing: 6255782.40
Hor. Datum: MGA2020-56
RL: 91.28 m
Ver. Datum: AHD
Surface: Pavement (Asphalt)

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ SOL ₆₄ VL ₂ L ₄ M ₂₀ H ₆₀ VH ₂₀₀ EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	50	Not Observed	D=0.83 MPa				SANDSTONE: medium to coarse grained, grey, thickly to thinly laminated at 0-10°, trace carbonaceous clasts.	FR				72.05 m: HB
	72.57		Is(50) A=2.2 MPa Is(50) D=1.8 MPa	19.0					100 (100) [100]			72.70 m: HB
	51		Is(50) A=2.1 MPa Is(50) D=1.8 MPa	73.0								73.35 m: HB 73.50 m: HB 73.66 m: HB 73.86 m: HB
	74.03			18.0								74.08 m: P, 10°, RF, UN
			Is(50) A=1.7 MPa Is(50) D=1.6 MPa	74.0					100 (100) [96]			74.39 m: HB
	52			17.0								74.74 m: HB 74.85 m: HB, clay 50mm
			Is(50)	75.0								75.06 m: HB
			A=1.7 MPa Is(50) D=1.7 MPa	16.0								
				76.0								
				15.0								
				77.0			Terminated at 75.17 m. Target depth.					
				14.0								
				78.0								
				13.0								
				79.0								
				12.0								
				80.0								

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A
Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	HWT	ES: 0.10-0.20 m PID=0.1 PPM	Not Observed	0.0			TOPSOIL - Silty SAND: fine grained, pale brown; with gravel, gravel is fine to medium, sub-angular to angular; trace clay; trace roots and root fibres; topsoil.	M		TOPSOIL
		ES: 0.50-0.60 m PID=0.1 PPM		0.66.0			FILL - Sandy CLAY: medium plasticity, pale yellow-brown mottled pale grey-brown; Sand, sand is fine to medium grained; trace gravel; trace roots and root fibres; trace ceramic fragments.			FILL
		D: 1.00-1.20 m ES: 1.00-1.10 m PID=0.1 PPM		1.0		CI	Sandy CLAY: medium plasticity, pale grey mottled orange-brown and yellow-brown; Sand, sand is fine to medium grained.	w ≈ PL	St	RESIDUAL SOIL
		SPT: 1.50-1.95 m 3.4.5 N=9 ES: 1.50-1.60 m PID=0.1 PPM PP=270, 230, 290 kPa ES: 2.00-2.10 m PID=0.1 PPM		2.0		SP	SAND: fine to medium grained, pale grey-white speckled dark grey.	M	VD	EXTREMELY WEATHERED MATERIAL
		SPT: 3.00-3.04 m 30/40 mm N=R ES: 3.00-3.10 m PID=0.3 PPM		3.0			SANDSTONE: medium grained, pale grey, very low strength. <i>Log continued on next page.</i>			HAWKESBURY SANDSTONE
				3.63.0						
				4.0						
				62.0						
				5.0						
				61.0						
				6.0						
				60.0						
				7.0						
				59.0						
				6.0						

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A
Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: ● D: ○ UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0							
					66.0							
					1.0							
					65.0							
					2.0							
					64.0							
					3.0							
							Log continued from previous page.					
	1 3.20				63.0		SANDSTONE: medium grained, pale grey, indistinct bedding.	HW	100 (100) [101]			HAWKESBURY SANDSTONE 3.05-4.00 m: MB, x 12
	2		Is(50) A=0.06 MPa Is(50) D=0.05 MPa		4.0		from 3.55 m: becoming iron-stained, yellow-brown and red-brown occasionally pale grey, occasional clay seams up to 80 mm thick	HW to MW				4.00-4.12 m: P, 3-5°, RF, CT clay, 60 mm spacing, x 2 4.21-4.50 m: EW, RF, clay, 20 to 80 mm aperture, 140 mm spacing, x 2
	4.70		Is(50) A=0.81 MPa Is(50) D=0.47 MPa C: 4.82-5.00 m		5.0			MW to SW	100 (97) [97]			4.57 m: P, 0°, RF, UN, CN 4.65-5.00 m: HB, x 2
	3				61.0		from 5.2 m: becoming medium grained, pale grey with dark grey laminations, very thinly cross-bedded at 10-20°, occasional shale inclusions up to 50 mm thick					5.03-5.25 m: MB, x 3
	5.70		Is(50) A=1.24 MPa Is(50) D=0.99 MPa		6.0							5.35 m: EW, RF, clay, 25 mm aperture
	4				6.0			SW to FR	100 (100) [100]			5.60-7.00 m: MB, x 5
	6.20				6.0							
	5		Is(50) A=0.64 MPa Is(50) D=1.1 MPa		7.0							
			Is(50) A=1.19 MPa Is(50) D=0.75 MPa		59.0			FR	100 (100) [100]			
					6.0							

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A
Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: * Is(50) D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	9.12		Is(50) A=0.63 MPa Is(50) D=1.6 MPa	58.0			SANDSTONE: medium grained, pale grey with dark grey laminations, very thinly cross-bedded at 10-20°, occasional shale inclusions, up to 50 mm thick.					
			Is(50) A=0.91 MPa Is(50) D=1.12 MPa	57.0			from 9.3 to 10.1 m: occasional shale gravel inclusions, trace carbonaceous laminations		100 (100) [97]			9.00-10.00 m: MB, x 2
			Is(50) A=1.42 MPa Is(50) D=1.61 MPa	56.0			from 10.11 to 10.18 m: conglomeratic sandstone, with shale clast inclusions					10.11-10.18 m: J, 10-20°, RF, SN, 40 mm spacing, x 2, shale clast
			Is(50) A=0.88 MPa Is(50) D=0.88 MPa	55.0			from 10.5 to 10.65 m: indistinct bedding					11.00-15.00 m: MB, x 7
			Is(50) A=1.17 MPa Is(50) D=1.2 MPa	54.0				FR	100 (100) [100]			
			Is(50) A=1.02 MPa Is(50) D=0.97 MPa	53.0								
			C: 14.27-14.42 m	52.0								
			Is(50) A=1.45 MPa Is(50) D=0.87 MPa	51.0					100 (100) [100]			15.00-18.00 m: MB, x 10
			Is(50) A=1.39 MPa Is(50) D=0.74 MPa	50.0								

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A

Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil Class: VL L M H VH 2 2 2 2 2 6 6 6 6 6 200 200 200 200 200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	8	Not Observed	Is(50) A=1.16 MPa Is(50) D=1.25 MPa	50.0		SANDSTONE: medium grained, pale grey with dark grey laminations, very thinly cross-bedded at 10-20°, occasional shale inclusions, up to 50 mm thick.						
			Is(50) A=1.33 MPa Is(50) D=1.12 MPa	17.0								
	Is(50) A=1.57 MPa Is(50) D=1.37 MPa		49.0		18.10 m: P, 5-7°, RF, UN, CN							
	Is(50) A=1.8 MPa Is(50) D=1.39 MPa		18.0		18.63-20.00 m: MB, x 13							
	Is(50) A=1.52 MPa Is(50) D=1.85 MPa		48.0		20.00-20.60 m: MB, x 5							
	Is(50) A=1.58 MPa Is(50) D=1.35 MPa		19.0		20.64 m: EW, RF, SN clay, 30 mm aperture							
	Is(50) A=2.12 MPa Is(50) D=1.66 MPa		47.0		21.00-21.16 m: MB, x 4							
	Is(50) A=1.65 MPa Is(50) D=1.51 MPa		20.0		21.20 m: J, 20°, RF, SN							
	Is(50) A=1.65 MPa Is(50) D=1.51 MPa		46.0		22.00-23.00 m: MB, x 4							
	Is(50) A=1.65 MPa Is(50) D=1.51 MPa		21.0		23.00-25.00 m: MB, x 11							
Is(50) A=1.65 MPa Is(50) D=1.51 MPa	45.0											
Is(50) A=1.65 MPa Is(50) D=1.51 MPa	22.0											
Is(50) A=1.65 MPa Is(50) D=1.51 MPa	44.0											
Is(50) A=1.65 MPa Is(50) D=1.51 MPa	23.0											
Is(50) A=1.65 MPa Is(50) D=1.51 MPa	43.0											

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A

Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (ROD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: ○ UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)																								
HQ3	24.20	Not Observed	Is(50) A=1.49 MPa Is(50) D=1.4 MPa	42.0		SANDSTONE: medium grained, pale grey with dark grey laminations, very thinly cross-bedded at 10-20°, occasional shale inclusions, up to 50 mm thick.	FR	100 (100) [97]			25.00-25.53 m: MB, x 4																									
	Is(50) A=0.58 MPa Is(50) D=0.4 MPa											41.0		SHALE: dark grey and occasionally grey, thinly laminated, sub-horizontal laminations at 0-3°. from 24.7 to 25.6 m: with sandstone laminations; sandstone is fine grained, pale grey.	100 (100) [100]		25.80 m: J, 15-20°, RF, ST, EW, 10 mm aperture 25.88-30.00 m: MB, x 20																			
																		Is(50) A=2.8 MPa Is(50) D=0.77 MPa	40.0		LAMINITE: fine grained, SANDSTONE (60%), fine grained, pale grey, interlaminated at 0-5° with SHALE (40%), dark grey.	100 (100) [100]		30.00-34.00 m: MB, x 16												
																									Is(50) A=2.97 MPa Is(50) D=0.77 MPa	38.0		27.0								
																															Is(50) A=2.21 MPa Is(50) D=0.6 MPa	29.0		36.0		
			Is(50) A=3.65 MPa Is(50) D=1.09 MPa C: 30.70-31.00 m	36.0		35.0																														
	Is(50) A=4.26 MPa Is(50) D=1.52 MPa									35.0																										

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A

Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ SOIL 0-6 VL 2 L 6 M 20 H 60 VH 200 EH	DEFECT SPACING (mm) 20 60 200 600 2000	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	33.19	Not Observed	Is(50) A=2.84 MPa Is(50) D=2.56 MPa	34.0		LAMINITE: fine grained, SANDSTONE (60%), fine grained, pale grey, interlaminated at 0-5° with SHALE (40%), dark grey. SANDSTONE: medium grained, pale grey occasionally stained pale brown, bedded at 0-20°.	FR	100 (100) [99]			34.40-35.00 m: EW, RF, clay, 20 mm aperture, 500 mm spacing, x 2 35.00-40.00 m: MB, x 20	
			Is(50) A=2.44 MPa Is(50) D=2.09 MPa	33.0								
			Is(50) A=1.42 MPa Is(50) D=1.5 MPa	33.0								
			Is(50) A=1.29 MPa Is(50) D=1.07 MPa	32.0								
			C: 35.00-35.10 m	35.0								
	36.20		Is(50) A=1.69 MPa Is(50) D=1.47 MPa	31.0								
			Is(50) A=1.85 MPa Is(50) D=1.66 MPa	30.0								
			Is(50) A=1.78 MPa Is(50) D=1.68 MPa	29.0								
			Is(50) A=1.39 MPa Is(50) D=1.46 MPa	28.0								
			Is(50) A=1.47 MPa Is(50) D=1.69 MPa	27.0								
39.21								100 (100) [100]				

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A
Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	SOIL ¹ ₂ VL ² L ³ M ⁴ H ⁵ VH ⁶ EH ⁷	INFERRED STRENGTH 20 x Is(50) A: ● D: ○ UCS: □ ⁸	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	16		Is(50) A=1.6 MPa Is(50) D=1.54 MPa	26.0			SANDSTONE: medium grained, pale grey occasionally stained pale brown, bedded at 0-20°.						40.00-45.00 m: MB, x 17
	42.20		Is(50) A=1.72 MPa Is(50) D=1.85 MPa	41.0									
			Is(50) A=1.54 MPa Is(50) D=1.7 MPa	25.0									
			Is(50) A=1.75 MPa Is(50) D=1.39 MPa	42.0			from 41.52 to 42.13 m: sandstone is fine grained, grey; trace gravel inclusions		100 (100) [100]				
			Is(50) A=2.38 MPa Is(50) D=1.81 MPa C: 44.58-44.97 m	24.0									
	45.20		Is(50) A=2.12 MPa Is(50) D=1.91 MPa	43.0									
			Is(50) A=1.75 MPa Is(50) D=1.99 MPa Is(50) A=1.74 MPa Is(50) D=1.82 MPa	23.0									
			Is(50) A=2.89 MPa Is(50) D=2.53 MPa	44.0									
				22.0					100 (100) [100]				45.00-49.67 m: MB, x 18
				45.0									
				21.0									
				46.0									
				20.0									
				47.0									
				19.0									
				48.0									

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
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Total Depth: 70.35 m
Hole Diameter: 120-90 mm
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Easting: 334516.80
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Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x IS ₍₅₀₎ A: • D: o UCS: □ L: 2 4 6 8 10 12 14 16 18 20 M: 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 H: 100 120 140 160 180 200 VH: 200 250 300 350 400 450 500 550 600 650 700 750 800 850 900 950 1000 EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	48.21	Not Observed	Is(50) A=1.7 MPa Is(50) D=1.1 MPa	18.0			SANDSTONE: medium grained, pale grey occasionally stained pale brown, bedded at 0-20°.	FR	100 (100) [100]			
	19		Is(50) A=2.72 MPa Is(50) D=2.05 MPa	17.0			from 48.8 to 50.75 m: sandstone is fine to coarse grained, pale grey speckled grey; with carbonaceous laminations					
	50.32		Is(50) A=1.93 MPa Is(50) D=1.71 MPa Is(50) A=1.43 MPa Is(50) D=1.93 MPa	16.0					100 (100) [100]			49.90 m: P, 0-3°, RF, UN, SN, 860 mm spacing, x 2
	20		Is(50) A=2.08 MPa Is(50) D=1.86 MPa Is(50) A=1.59 MPa Is(50) D=1.58 MPa	15.0								50.32 m: MB
	53.32		Is(50) A=1.63 MPa Is(50) D=1.87 MPa	14.0					100 (99) [99]			50.75 m: P, 5°, SM, SN, shale clasts 50.83-54.03 m: MB, x 12
	21		Is(50) A=1.9 MPa Is(50) D=1.62 MPa	13.0								
	54.23		Is(50) A=1.8 MPa Is(50) D=1.74 MPa	12.0					100 (100) [100]			54.10 m: EW, RF, SN clay, 15 mm aperture
	22											54.50-55.00 m: MB, x 4
	55.00								100 (100) [100]			55.00-55.60 m: MB, x 9
	23		Is(50) A=2.02 MPa Is(50) D=1.85 MPa	11.0								

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A

Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: • D: o UCS: □ Soil: VL L M H VH EH	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	56.00	Not Observed	Is(50) A=1.51 MPa Is(50) D=1.51 MPa	10.0	57.0		SANDSTONE: medium grained, pale grey occasionally stained pale brown, bedded at 0-20°. SHALE: dark grey, bedded at 0-5°. SANDSTONE: medium grained, pale grey stained pale brown, bedded at 0-20°, occasionally cross-bedded, occasional shale laminations.	FR	100 (100) [100]			56.84 m: P, 0-3°, RF, CN shale clasts inclusions 57.00-59.37 m: MB, x 10
	24											
	57.00											
	25											
	58.00											
	26											
	59.00											
	27											
	60.00											
	28											
	61.00	Not Observed	Is(50) A=2.38 MPa Is(50) D=1.6 MPa	61.0	62.0		from 60.0 - 63.5 m: indistinct bedding, occasional dark grey laminations	FR	100 (100) [100]			59.90 m: EW, 15-30°, RF, SN clay, 10 mm aperture 60.00-65.00 m: MB, x 15
	29											
	62.00											
	30											
	63.25											
	3.0											
	64.0											

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: Alex To
Start Date: 27/10/25
Checked by: J Riley
End Date: 03/11/25

Driller: Terratest
Drill Rig: Comacchio Geo 405
Total Depth: 70.35 m
Hole Diameter: 120-90 mm
Inclination: 90°
Bearing: N/A

Easting: 334516.80
Northing: 6255598.40
Hor. Datum: MGA2020-56
RL: 66.391 m
Ver. Datum: AHD
Surface: Grass

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) (RQD) (%)	INFERRED STRENGTH 20 x Is(50) A: * D: o UCS: □ L: 2 M: 20 H: 60 VH: 200 VL: 2 L: 2 M: 20 H: 60 VH: 200	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
HQ3	31	Not Observed	Is(50) A=1.81 MPa Is(50) D=1.35 MPa	66.25	65.0		SANDSTONE: medium grained, pale grey stained pale brown, bedded at 0-20°, occasionally cross-bedded, occasional shale laminations.	FR	100 (100) [100]			65.00-70.00 m: MB, x 14
	32		C: 66.13-66.26 m Is(50) A=1.2 MPa Is(50) D=0.8 MPa Is(50) A=1.29 MPa Is(50) D=0.8 MPa	69.25	67.0		CONGLOMERATE: fine grained, pale grey speckled grey white and dark grey, indistinctly bedded, sandstone matrix. SANDSTONE: medium grained, dark grey and grey, thinly laminated at 15°. SILTSTONE: medium grained, dark grey and grey, indistinct bedding, bioturbidity visible.					
	33		Is(50) A=1.39 MPa Is(50) D=1.48 MPa Is(50) A=2 MPa Is(50) D=1.81 MPa		70.0		SANDSTONE: medium grained, pale grey speckled dark grey, indistinct bedding. from 69.4 to 69.48 m: siltstone lamination, dark grey, indistinct bedding from 69.52 to 69.76 m: siltstone rip up clasts, 2-40 mm thick from 70.16 to 70.19 m: shale lamination, dark grey		100 (100) [100]			70.00-70.35 m: MB, x 3
					72.0		Terminated at 70.35 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: A Harrison
Start Date: 04/11/25
Checked by: J Riley
End Date: 04/11/25

Driller: Terratest
Drill Rig: MC-T200
Total Depth: 6 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A
Easting: 334522.40
Northing: 6255616.90
Hor. Datum: MGA2020-56
RL: 66.185 m
Ver. Datum: AHD
Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA		PID=0.7 PPM		66.0	0.0			FILL - Silty SAND: fine to medium grained, brown; trace clay; trace gravel, fine to medium grained (Sandstone); trace root fibres; trace bricks, glass.			FILL
		ES: 0.50-0.60 m PID=1.2 PPM			1.0				D		
		ES: 1.00-1.10 m PID=1.1 PPM		65.0				SANDSTONE: fine to medium grained, orange-brown mottled red-brown, soil strength.			EXTREMELY WEATHERED MATERIAL
								at 1.60 m: becoming red-brown			
					2.0						
					64.0						
					3.0						
					63.0						
					4.0						
					62.0						
					5.0						
					61.0						
					6.0			Terminated at 6.00 m.			
					60.0						
					7.0						
					59.0						
					6.0						

Client:

Sydney Water

Project:

Willoughby Submain Amplification Project

Location:

ANZAC Park, Cammeray, 2062

Project No:

60764076

Logged by:

A Harrison

Checked by:

J Riley

Start Date:

04/11/25

End Date:

04/11/25

Driller:

Terratest

Drill Rig:

MC-T200

Total Depth:

1.5 m

Hole Diameter:

96 mm

Inclination:

90°

Bearing:

N/A

Easting:

334507.50

Northing:

6255616.00

Hor. Datum:

MGA2020-56

RL:

66.831 m

Ver. Datum:

AHD

Surface:

Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	ES: 0.10-0.20 m PID=0.8 PPM	Not Observed	66.0	0.0			FILL - Silty SAND: fine to medium grained, brown; trace clay; trace gravel, fine to medium grained (Sandstone); trace root fibres.	D		FILL
AD/T		ES: 0.50-0.60 m PID=1.2 PPM			1.0			SANDSTONE: fine to medium grained, red-brown, soil strength. at 1.20 m: becoming light brown			EXTREMELY WEATHERED MATERIAL
				65.0	2.0			Terminated at 1.50 m.			
				64.0	3.0						
				63.0	4.0						
				62.0	5.0						
				61.0	6.0						
				60.0	7.0						
				59.0							
				58.0							

Client: Sydney Water
Project: Willoughby Submain Amplification Project
Location: ANZAC Park, Cammeray, 2062

Project No: 60764076
Logged by: A Harrison
Start Date: 04/11/25
Checked by: J Riley
End Date: 04/11/25

Driller: Terratest
Drill Rig: MC-T200
Total Depth: 2.5 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A
Easting: 332747.90
Northing: 6255596.10
Hor. Datum: MGA2020-56
RL: 67.23 m
Ver. Datum: AHD
Surface: Grass

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA		ES: 0.10-0.20 m PID=1 PPM		67.0	0.0			FILL - Silty SAND: fine to medium grained, brown; trace clay; trace gravel, fine to medium grained (Sandstone); trace root fibres.			FILL
		ES: 0.50-0.60 m PID=0.7 PPM						FILL - SAND: fine to medium grained, pale brown; trace clay; trace gravel, fine to medium grained (Sandstone); trace root fibres.			
AD/T	U	ES: 1.00-1.10 m PID=1.2 PPM	Not Observed	66.0	1.0		CL	Sandy CLAY: low plasticity, yellow-brown; Sand, fine to medium grained.	D		RESIDUAL SOIL
		ES: 1.50-1.60 m PID=1.2 PPM						Sandy CLAY: medium plasticity, orange-brown; Sand, fine to medium grained.			
		ES: 2.00-2.10 m PID=1.0 PPM		65.0	2.0		CI	at 2.00 m: becoming orange-brown mottled red-brown			
								Terminated at 2.50 m.			
					3.0						
				64.0							
					4.0						
				63.0							
					5.0						
				62.0							
					6.0						
				61.0							
					7.0						
				60.0							
					8.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.